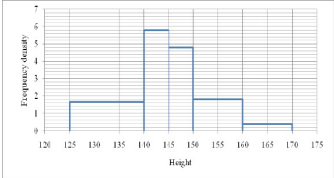


Question			Answer/Indicative content	Marks	Part marks and guidance																									
1		i	$4 + \frac{1}{2} \text{ of } 18 = 4 + 9 = 13$	M1	For $\frac{1}{2}$ of 18																									
		i		A1	cao Examiner's Comments On the whole, this question was answered well. The most common incorrect answer was 22, which was seen fairly frequently. A small minority of candidates wrote $9 + 4$, but then calculated incorrectly (both 11 and 12 seen).	13/100 gets M1A0																								
		ii	(Median) = 50.5 th value	M1	For 50.5 seen	SC2 for use of 50 th value leading to Est = $140 + (25/29 \times 5) = 144.3$ (SC1 if over-specified)																								
		ii	Est = $140 + \left(\frac{25.5}{29}\right) \times 5$ or = $140 + \left(\frac{50.5 - 25}{54 - 25}\right) \times 5$	M1	For attempt to find this value	or Est = $145 - \left(\frac{3.5}{29}\right) \times 5 = 144.4$																								
		ii	= 144.4	A1	Examiner's Comments Only about 10% of candidates produced totally correct answers. Many scored SC2 for finding the 50 th , rather than 50.5 th , value. Those that did state that they were looking for 50.5 th value often just gave the mid-value, rather than using interpolation. Many candidates lost a mark due to over-specification.	NB no marks for mean = 144.35 NB Watch for over-specification																								
		iii	<table><tr><th>Height</th><th>Frequency</th><th>Group width</th><th>Frequency density</th></tr><tr><td>$125 \leq x \leq 140$</td><td>25</td><td>15</td><td>1.67</td></tr><tr><td>$140 < x \leq 145$</td><td>29</td><td>5</td><td>5.80</td></tr><tr><td>$145 < x \leq 150$</td><td>24</td><td>5</td><td>4.80</td></tr><tr><td>$150 < x \leq 160$</td><td>18</td><td>10</td><td>1.80</td></tr><tr><td>$160 < x \leq 170$</td><td>4</td><td>10</td><td>0.40</td></tr></table> **	Height	Frequency	Group width	Frequency density	$125 \leq x \leq 140$	25	15	1.67	$140 < x \leq 145$	29	5	5.80	$145 < x \leq 150$	24	5	4.80	$150 < x \leq 160$	18	10	1.80	$160 < x \leq 170$	4	10	0.40	M1	For fd's - at least 3 correct	M1 can be also be gained from freq per 10 – 16.7, 58, 48, 18, 4 (at least 3 correct) or freq per 5 – 8.35, 29, 24, 9, 2 for all correct.
Height	Frequency	Group width	Frequency density																											
$125 \leq x \leq 140$	25	15	1.67																											
$140 < x \leq 145$	29	5	5.80																											
$145 < x \leq 150$	24	5	4.80																											
$150 < x \leq 160$	18	10	1.80																											
$160 < x \leq 170$	4	10	0.40																											

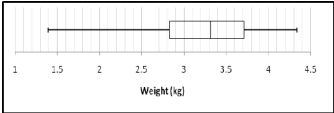
Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii		A1	Accept any suitable unit for fd such as eg freq per cm. correct to at least one dp allow 1.66 but not 1.6 for first fd	If fd not explicitly given, M1 A1 can be gained from all heights correct (within one square) on histogram (and M1A0 if at least 3 correct)
		iii		G1	linear scales on both axes and label on vertical axis	Linear scale and label on vertical axis IN RELATION to first M1 mark ie fd or frequency density or if relevant freq / 10, etc (NOT eg fd / 10). However allow scale given as fd × 10, or similar Accept f/w or f/cw (freq / width or freq / class width) Can also be gained from an accurate key G0 if correct label but not fd's.
		iii		W1	width of bars	Must be drawn at 125, 140 etc NOT 124.5 or 125.5 etc NO GAPS ALLOWED Must have linear scale. No inequality labels on their own such as $125 \leq S < 140$, etc but allow if a clear horizontal linear scale is also given. Ignore horizontal label.

Question			Answer/Indicative content	Marks	Part marks and guidance													
		iii		H1	height of bars Examiner's Comments The histogram was generally completed rather better than in previous years. Most candidates were able to calculate frequency densities correctly, and they also usually labelled the axes correctly. A fairly common error was to round the first frequency density down to 1.6 rather than to 1.7. Some made errors with careless drawing of bars, making slips with incorrect heights.	Height of bars – must be linear vertical scale. FT of heights dep on at least 3 heights correct and all must agree with their fds If fds not given and at least 3 heights correct then max M1A0G1W1H0 Allow restart with correct heights if given fd wrong (for last three marks only)												
		iv	4 boys	3														
		iv	0.6×15	M1	For 0.6×15	Or $45 \times 0.2 = 9$ (number of squares and 0.2 per square)												
		iv	= 9 girls	A1	For 9 girls													
		iv	So 5 more girls	A1	cao													
					Examiner's Comments Roughly 90% of candidates scored full marks here.													
		v	Frequencies and midpoints for girls are <table><tr><td>Height</td><td>132.5</td><td>142.5</td><td>147.5</td><td>155</td><td>167.5</td></tr><tr><td>Frequency</td><td>18</td><td>23</td><td>31</td><td>19</td><td>9</td></tr></table>	Height	132.5	142.5	147.5	155	167.5	Frequency	18	23	31	19	9	B1	For at least three frequencies correct	
Height	132.5	142.5	147.5	155	167.5													
Frequency	18	23	31	19	9													
		v		B1	At least three midpoints correct	No further marks if not using midpoints												
		v	So mean =	M1	For attempt at Σxf	For sight of at least 3 xf pairs												
		v	$\frac{(132.5 \times 18) + (142.5 \times 23) + (147.5 \times 31) + (155 \times 19) + (167.5 \times 9)}{100}$ $= \frac{(2385) + (3277.5) + (4572.5) + (2945) + (1507.5)}{100}$	M1* Dep on M1	For division by 100													

Question			Answer/Indicative content	Marks	Part marks and guidance	
		v		A1	Cao NB Watch for over-specification Examiner's Comments Many candidates did everything correctly but gave the final answer as 146.875 or 146.88 thus losing the final mark for over-specifying. The scheme allowed for a slip in both the frequencies and the mid-points and candidates were still able to gain 4 marks. The most common error was giving the final mid-point as 165 rather than 167.5.	Allow answer 146.9 or 147 but not 150 NB Accept answers seen without working (from calculator) Use of 'not quite right' midpoints such as 132.49 or 132.51 etc can get B1B0M1M1A0
			Total	18		

Question			Answer/Indicative content	Marks	Part marks and guidance	
2		i	Positive	B1	CAO Examiner's Comments Approximately 95% of candidates scored this mark.	
		ii	Mean = 5.064 allow 5.1 with working 126.6/25 or 5.06 without	B1		
		ii	SD = 1.324 allow 1.3 with working or 1.32 without	B2	Allow B1 for RMSD = 1.297 or var = 1.753 or MSD = 1.683 Examiner's Comments There were many fully correct answers. Most used the relevant formulas rather than using the built in functions on their calculators. A few candidates found the variance or the rmsd, and these gained a method mark. The most common error was not to use the key and thus get answers ten times too high. This error was severely penalised, but full marks were allowed for a follow through.	Also allow B1 for $S_{xx} = 42.08$ or for $\Sigma x^2 = 683$ SC1 for both mean = 50.64 and SD = 13.24 (even if over-specified)
		iii	$\bar{x} - 2s = 5.064 - 2 \times 1.324 = 2.416$	B1FT	FT their mean and sd	For use of quartiles and IQR $Q_1 = 3.95$; $Q_3 = 6.0$; IQR = 2.05 $3.95 - 1.5(2.05)$ gets M1 Allow other sensible definitions of quartiles
		iii	$\bar{x} + 2s = 5.064 + 2 \times 1.324 = 7.712$	M1	for $\bar{x} + 2s$ but withhold final E mark if their limits mean that there are no outliers.	$6.0 + 1.5(2.05)$ gets M1
		iii		A1FT	For upper limit	Limits 0.875 and 9.075

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	So there is an outlier.	E1	Incorrect statement such as 7.6 and 8.1 are outliers gets E0 Do not award E1 if calculation error in upper limit Examiner's Comments The limits for outliers were widely known and correctly used by most candidates. Even those with incorrect mean and standard deviation were able to gain 3 or all 4 marks if they followed through correctly. Some candidates used the quartiles method, despite often having got part (ii) correct, but some of these made errors, losing some if not all of the marks.	So there are no outliers NB do not penalise over-specification here as not the final answer but just used for comparison. FT from SC1
			Total	8		

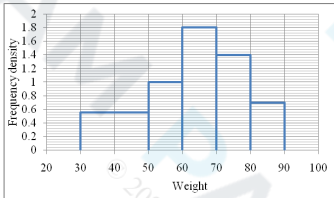
Question			Answer/Indicative content	Marks	Part marks and guidance	
3		i	Median = 3.32 kg	B1		
		i	Q1 (= 6.5th value) = 2.83 Q3 (= 19.5th value) = 3.71	B1	For Q1 or Q3	For Q1 allow 2.82 to 2.84
		i	Inter-quartile range = $3.71 - 2.83 = 0.88$	B1	For IQR dep on both quartiles correct <u>Examiner's Comments</u> Most candidates successfully found the median, although instead of the 13th value some found average of the 12 th and 13 th values. However, candidates were less successful in finding the interquartile range. The lower quartile was usually found correctly, but the upper quartile was more frequently wrong, with an answer of 3.665 being the most common error. Occasionally candidates did not subtract to find the interquartile range, but instead some found the midpoint of their quartiles.	For Q3 allow 3.70 to 3.72 If no quartiles given allow B0B1 for IQR in range 0.86 to 0.90
		ii		G1	For reasonably linear scale shown.	Dep on attempt at box and whisker plot with at least a box and one whisker. Condone lack of label.
		ii		G1	For boxes in approximately correct positions, with median just to right of centre	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii		G1	For whiskers in approximately correct positions in proportion to the box FT their median and quartiles if sensible – guidance above is only for correct values <u>Examiner's Comments</u> The response to this question was very disappointing. Perhaps because they were faced with a blank space rather than graph paper, most candidates thought that accuracy was not required. Very few had a scale and some of those that did failed to make it linear. Some candidates simply sketched a box and whisker plot and then labelled the diagram with the relevant values. This did not gain marks as the question clearly instructs candidates to 'Draw a box and whisker plot...'. It seems likely that many candidates either did not have, or did not think to use a ruler. Far too many freehand diagrams were seen, with the sizes of the box and whiskers and the position of the median not in proportion.	Do not award unless RH whisker significantly shorter than LH whisker Allow LH whisker going to 2.5 and outlier marked at 1.39
		iii	Lower limit $2.83 - (1.5 \times 0.88) = 1.51$	B1	For 1.51 FT	Any use of median $\pm 1.5 \times$ IQR scores B0 B0 E0 No marks for ± 2 or $3 \times$ IQR In this part FT their values from (i) or (ii) if sensibly obtained but not from location ie 6.5, 19.5

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	Upper limit $3.71 + (1.5 \times 0.88) = 5.03$	B1	For 5.03 FT	Do not penalise over-specification as not the final answer
		iii	Exactly one baby weighs less than 1.51 kg and none weigh over 5.03 kg so there is exactly one outlier.	E1*	Dep on their 1.51 and 5.03	Do not allow unless FT leads to upper limit above 4.34 and lower limit between 1.39 and 2.50
		iii	'Nothing to suggest that this baby is not a genuine data value so she should not be excluded' or 'This baby is premature and therefore should be excluded'.	E1* Dep	Any sensible comment in context Examiner's Comments Many candidates correctly found the upper and lower limits for the outliers. The most common misconception was that outliers were calculated using $\text{median} \pm 1.5 \times \text{IQR}$, although many other errors were also seen. A few candidates attempted to use the mean and standard deviation, and if they got both of these correct, full marks were available, but unfortunately one or other of the two statistics was usually incorrect. It was necessary to check both limits to show that there was only one outlier, but some candidates ignored the upper limit. Many candidates failed to give an explanation in context regarding the outlier, though those that did often made a valid point about premature babies.	For use of mean $\pm 2\text{sd}$ allow B1 For $3.27 + 2 \times 0.62 = 4.51$ B1 For $3.27 - 2 \times 0.62 = 2.03$ Then E1E1 as per scheme
		iv	Median = 3.5 kg	B1		
		iv	Q1 = 50th value = 3.12 Q3 = 150th value = 3.84	B1	For Q1 or Q3	For Q1 allow 3.11 to 3.13 For Q3 allow 3.83 to 3.85

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv	Inter-quartile range = $3.84 - 3.12 = 0.72$	B1	For IQR FT their quartiles <u>Examiner's Comments</u> As in part (i), the median was usually found correctly, but some candidates lost a mark due to inaccurate reading of the scales in finding the quartiles.	Dep on both quartiles correct If no quartiles given allow B0B1 for IQR in range 0.70 to 0.74
		v	Female babies have lower weight than male babies on the whole	E1	Allow 'on average' or	Do not allow lower median
		v		FT	similar in place of 'on the whole'	
		v	Female babies have higher weight variation than male babies	E1	Allow 'more spread' or	Do not allow higher IQR, but SC1 for
		v		FT	similar but not 'higher range' Condone less consistent <u>Examiner's Comments</u> Only about one third of candidates scored both marks. Credit was given to those candidates who could only compare medians and interquartile ranges without an explanation of what they meant. Candidates who just said 'boys are heavier' failed to get credit without a comment such as 'generally' or 'on average' or 'tend to be'. Similarly 'more consistent' or 'vary less' or 'less spread' gained credit for interquartile range – 'smaller range' was not awarded credit.	both lower median and higher IQR, making clear which is which

Question			Answer/Indicative content	Marks	Part marks and guidance	
		vi	Male babies must weigh more than 4.34 kg			
		vi	Approx 10 male babies weigh more than this.	M1*	For 10 or 9 or 8	Or 200 – 190, 200 –191 or 200 –192
		vi	$\text{Probability} = \frac{10}{200} \times \frac{9}{199} = \frac{90}{39800} = \frac{9}{3980} = 0.00226$ $\text{or } \frac{9}{200} \times \frac{8}{199} = \frac{72}{39800} = 0.00181$	M1* dep	For first fraction multiplied by any other different fraction (Not a binomial probability)	Allow any of these answers For spurious factors, eg 2 × correct answer allow M1M1A0
		vi	$\text{or } \frac{8}{200} \times \frac{7}{199} = \frac{56}{39800} = \frac{7}{4975} = 0.00141$	A1	CAO Allow their answer to min of 2 sf <u>Examiner's Comments</u> This part discriminated very well between the higher-scoring candidates. Many candidates realised that approximately 10 male babies weighed more than 4.34 kg. Unfortunately many then did not know how to proceed, often squaring 0.05 (10/200) rather than multiplying by 9/199. Those candidates who misread the scale but knew how to proceed could gain a Special Case mark. A significant number of candidates missed out this part altogether.	SC1 for $n/200 \times (n-1)/199$ NOTE RE OVER-SPECIFICATION OF ANSWERS If answers are grossly over-specified, deduct the final answer mark in every case. Probabilities should also be rounded to a sensible degree of accuracy. In general final non probability answers should not be given to more than 4 significant figures. Allow probabilities given to 5 sig fig. PLEASE HIGHLIGHT ANY OVER-SPECIFICATION Please note that there are no G or E marks in scoris, so use B instead
			Total	18		

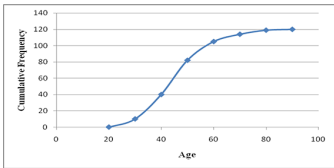
Question			Answer/Indicative content	Marks	Part marks and guidance																									
4		i	<table><thead><tr><th>Weight</th><th>Frequency</th><th>Group Width</th><th>Frequency density</th></tr></thead><tbody><tr><td>$30 \leq w < 50$</td><td>11</td><td>20</td><td>0.55</td></tr><tr><td>$50 \leq w < 60$</td><td>10</td><td>10</td><td>1</td></tr><tr><td>$60 \leq w < 70$</td><td>18</td><td>10</td><td>1.8</td></tr><tr><td>$70 \leq w < 80$</td><td>14</td><td>10</td><td>1.4</td></tr><tr><td>$80 \leq w < 90$</td><td>7</td><td>10</td><td>0.7</td></tr></tbody></table>	Weight	Frequency	Group Width	Frequency density	$30 \leq w < 50$	11	20	0.55	$50 \leq w < 60$	10	10	1	$60 \leq w < 70$	18	10	1.8	$70 \leq w < 80$	14	10	1.4	$80 \leq w < 90$	7	10	0.7	M1	For fd's – at least 3 correct Accept any suitable unit for fd such as eg freq per 10g.	M1 can be also be gained from freq per 10 – 5.5, 10, 18, 14, 7 (at least 3 correct) or similar. If fd not explicitly given, M1
	Weight	Frequency	Group Width	Frequency density																										
	$30 \leq w < 50$	11	20	0.55																										
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$80 \leq w < 90$	7	10	0.7																											
	i			A1		A1 can be gained from all heights correct (within half a square) on histogram (and M1A0 if at least 3 correct) Linear scale and label on vertical axis IN RELATION to first M1 mark ie fd or frequency density or if relevant freq/10, etc (NOT eg fd/10).																								
	i		G1	linear scales on both axes and labels Vertical scale starting from zero (not broken – but can get final mark for heights if broken)	However allow scale given as $fd \times 10$, or similar. Accept f/w or f/cw (freq/width or freq/class width) Ignore horizontal label Can also be gained from an accurate key G0 if correct label but not fd's.																									
	i			G1	width of bars	Must be drawn at 30, 50 etc NOT 29.5 or 30.5 etc NO GAPS ALLOWED Must have linear scale. No inequality labels on their own such as $30 \leq W < 50$, $50 \leq W < 60$ etc but allow if 30, 50, 60 etc occur at the correct boundary position. See additional notes. Allow this mark even if not using fd's																								

Question			Answer/Indicative content	Marks	Part marks and guidance	
		i		G1	height of bars Examiner's Comments Most candidates found the frequency densities correctly. They usually then went on to draw the axes correctly although a few failed to start the frequency density scale at zero or to label the axes. A few candidates used inequalities on the horizontal axis, which attracted a penalty of one mark. The choice of scales on the vertical axis was not always ideal, and this left some candidates vulnerable to drawing the heights at incorrect positions. In particular the height of the first bar was frequently incorrectly plotted at 0.5 rather than 0.55.	Height of bars – must be linear vertical scale. FT of heights dep on at least 3 heights correct and all must agree with their fds If fds not given and at least 3 heights correct then max M1A0G1G1G0 Allow restart with correct heights if given fd wrong (for last three marks only)
		ii	Mean= $\frac{(40 \times 11) + (55 \times 10) + (65 \times 18) + (75 \times 14) + (85 \times 7)}{60} = \frac{3805}{60}$	M1	For midpoints Products are 440, 550, 1170, 1050, 595	For midpoints (at least 3 correct) No marks for mean or sd unless using midpoints

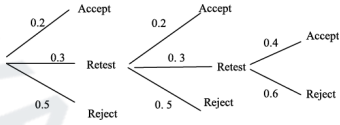
Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	= 63.4 (or 63.42)	A1	CAO (exact answer 63.41666...)	Answer must NOT be left as improper fraction as this is an estimate Accept correct answers for mean and sd from calculator even if eg wrong Sxx given
		ii	$\sum x^2 f = (40^2 \times 11) + (55^2 \times 10) + (65^2 \times 18) + (75^2 \times 11) + (85^2 \times 7)$ $= 253225$		Examiner's Comments The calculation of the mean of the grouped data was in most cases accurately performed using correct mid-points. The calculation of the standard deviation was less well executed. Whilst there were many correct solutions seen, some forgot to factor in the frequencies and worked with $\sum x^2$ rather than $\sum fx^2$. Over specification of either or both of the answers caused some candidates to lose one mark.	
		ii	$S_{xx} = 253225 - \frac{3805^2}{60} = 11924.6$	M1	For attempt at S_{xx} Should include sum of at least 3 correct multiples $fx^2 - \sum x^2/n$	Allow M1 for anything which rounds to 11900
		ii	$s = \sqrt{\frac{11924.6}{59}} = \sqrt{202.11} = 14.2$		At least 1dp required Use of mean 63.4 leading to answer of 14.29199.. with $S_{xx} = 12051.4$ gets full credit.	Allow SC1 for RMSD 14.1 (14.0976...) from calculator.
		ii		A1		Only penalise once in part (ii) for over specification, even if mean and standard deviation both over specified.
		ii			63.42 leads to 14.2014... Do not FT their incorrect mean (exact answer 14.2166...)	If using $(x - \bar{x})^2$ method, B2 if 14.2 or better (14.3 if use of 63.4), otherwise B0

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$\bar{x} - 2s = 63.4 - (2 \times 14.2) = 35$	M1	For either No marks in (iii) unless using $\bar{x} + 2s$ or $\bar{x} - 2s$	FT their positive mean and their positive sd / rmsd for M1A1. Only follow through numerical values, not variables such as s , so if a candidate does not find s but then writes here 'limit is $63.4 + 2 \times$ standard deviation', do NOT award M1
		iii	$\bar{x} + 2s = 63.4 + (2 \times 14.2) = 91.8$	A1	For both (FT)	Do not penalise for over-specification
		iii	So there are probably some outliers at the lower end, but none at the upper end	E1	Must include an element of doubt and must mention both ends Examiner's Comments Most candidates scored at least the first two marks. However many omitted the fact that there were definitely no outliers at the top end of the data and/or stated that there were definitely some outliers present at the bottom end, thus missing the final mark.	Must have correct limits to get this mark
		iv	Mean = $\frac{3624.5}{50} = 72.5\text{g}$ (or exact answer 72.49g)	B1	CAO Ignore units	
		iv	$S_{xx} = 265416 - \frac{3624.5^2}{50} = 2676$	M1	For S_{xx}	M1 for $265416 - 50 \times \text{their mean}^2$ BUT NOTE M0 if their $S_{xx} < 0$

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv	$s = \sqrt{\frac{2676}{49}} = \sqrt{54.61} = 7.39\text{g}$	A1	CAO ignore units Allow 7.4 but NOT 7.3 (unless RMSD with working) Examiner's Comments This was generally very well answered.	For s^2 of 54.6 (or better) allow M1A0 with or without working. For RMSD of 7.3 (or better) allow M1A0 provided working seen For RMSD ² of 53.5 (or better) allow M1A0 provided working seen
		v	Variety A have lower average than Variety B oe	E1	FT their means Do not condone lower central tendency or lower mean	Allow 'on the whole' or similar in place of 'average'.
		v	Variety A have higher variation than Variety B oe	E1	FT their sd Examiner's Comments For this type of question candidates should be taught to discuss 'average' and 'variation'. Simply stating for example that the mean of A is lower than the mean of B does not attract any credit.	Allow 'more spread' or similar but not 'higher range' or 'higher variance' Condone less consistent.
			Total	17		

Question			Answer/Indicative content	Marks	Part marks and guidance																			
5		i	<table><tr><td>Upper Bound</td><td>20</td><td>30</td><td>40</td><td>50</td><td>60</td><td>70</td><td>80</td><td>90</td></tr><tr><td>Cumulative Freq</td><td>0</td><td>10</td><td>40</td><td>82</td><td>105</td><td>114</td><td>119</td><td>120</td></tr></table>	Upper Bound	20	30	40	50	60	70	80	90	Cumulative Freq	0	10	40	82	105	114	119	120	B1	Cumulative frequencies All correct	May be implied from graph. Condone omission of 0 at this stage.
	Upper Bound	20	30	40	50	60	70	80	90															
	Cumulative Freq	0	10	40	82	105	114	119	120															
		i		G1	For plotted points (Provided plotted at correct UCB positions)	Plotted as (UCB, their cf). Ignore (20,0) at this stage. No midpoint or LCB plots. Plotted within ½ small square If cf not given then allow G1 for good attempt at cf. e.g. if they have 0,10,40,72,95,104,109,110																		
	i		G1	For joining points (within ½ a square)	For joining all of 'their points' (line or smooth curve) AND now including (20,0) Not for midpoint or LCB plots.																			
		i		G1	For scales	Linear horizontal scale. Allow if start at 30 (no inequality scales - Not even <20, <30, <40 ...) Linear vertical scale Allow full credit if axes reversed correctly																		

Question			Answer/Indicative content	Marks	Part marks and guidance	
		i		G1	For labels All marks dep on good attempt at cumulative frequency, but not cumulative fx's or other spurious values. Examiner's Comments Many candidates gained full credit. A common error which resulted in the loss of 2 marks was to plot the correct height but at mid-points. Only a few used the lower class boundaries. Some candidates drew cumulative frequency bars and a small number just plotted frequency against midpoints. Some candidates forgot to label their axes or more often omitted the word "cumulative" on their vertical axis.	Age or x and Cumulative frequency or just CF or similar but not just frequency or fd nor cumulative fd Mid-point or LCB plots may score first and last two marks Can get up to 3/5 for cum freq bars Lines of best fit could attract max 4 out of 5.
		ii	Median = 45	B1	Allow answers between 44 and 46 without checking curve. Otherwise check curve. No marks if not using diagram.	Based on 60th value ft their curve (not LCB's) Allow 40 for m.p. plot without checking graph B0 for interpolation If max value wrong (eg 110) FT their max value for all 3 marks
		ii	Q1 = 37 Q3 = 53	B1	For Q3 or Q1 Allow Q1 between 37 and 38 without checking Allow Q3 between 52 and 54 without checking	Based on 30th and 90th values ft their curve (not LCB's) Allow Q1 = 32; Q3 = 48 without checking graph

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	Inter-quartile range = $53 - 37 = 16$	B1	For IQR providing both Q1 and Q3 are correct	B0 for interpolation B2 for correct IQR from graph if quartiles not stated but indicated on graph Allow from mid-point plot Must be good attempt at cumulative frequency in part (i) to score any marks here Lines of best fit: B0 B0 B0 here. Also cumulative frequency bars: B0 B0 B0 here
		ii			Alternative version of tree diagram for Q2(i) Examiner's Comments This part was very well answered with many candidates picking up the follow through marks for correctly identifying the median and quartiles from their mid-point plotted graph.	
			Total	8		

Question			Answer/Indicative content	Marks	Part marks and guidance	
6		i	Inter-quartile range = $18.1 - 17.8 = 0.3$	B1		
		i	Lower limit $17.8 - (1.5 \times 0.3) (= 17.35)$	M1		FT their IQR for M marks only
		i	No outliers at lower end.	A1	dep on 17.35	Allow 'No values below 17.35 for first A1 Allow 'Lower limit = 17.35 so no outliers (at lower end)' Watch for use of median giving 17.45 which gets M0A0 You must be convinced that comments about no outliers refer to lower tail only.
		i	Upper limit $18.1 + (1.5 \times 0.3) (= 18.55)$	M1		
		i	(Max is 18.6) so at least one outlier at upper end.	A1	dep on 18.55	Allow 'At least one value above 18.55' for second A1 Allow 'any value above 18.55 is an outlier' so at least one outlier. Do not allow 'There MAY be one outlier' oe Condone 'one outlier' Condone 'there are outliers' Watch for use of median giving 18.35 which gets M0A0 You must be convinced that comments about some outliers refer to upper tail only.

Question			Answer/Indicative content	Marks	Part marks and guidance	
		i			Examiner's Comments Most candidates calculated the inter-quartile range and used it correctly to find the limits for outliers. However a few used the median instead of the quartiles to add to and subtract from $1.5 \times$ inter-quartile range. Many candidates neglected to comment on outliers at each end separately (only commenting if there were outliers overall).	
		ii	$P(A) = P(\text{All 3 have orange centres}) =$ $\frac{7}{20} \times \frac{6}{19} \times \frac{5}{18} = \frac{7}{228}$	M1	For $7/20 \times$	Allow final answer of 0.031 with working
		ii		M1	For product of correct three fractions Without extra terms	ALTERNATIVE SCHEME ⁷ $C_3/^{20}C_3$ $= 35/1140 = 7/228 = 0.0307$ M1 for either term in correct position in a fraction
		ii	$= 0.0307 \text{ (0.030702)}$	A1	CAO Allow full marks for fully simplified fractional answers	M1 for correct fraction A1 CAO
		ii	$P(B) = P(\text{All 3 have same centres}) =$ $\left(\frac{7}{20} \times \frac{6}{19} \times \frac{5}{18}\right) + \left(\frac{6}{20} \times \frac{5}{19} \times \frac{4}{18}\right) +$ $\left(\frac{4}{20} \times \frac{3}{19} \times \frac{2}{18}\right) + \left(\frac{3}{20} \times \frac{2}{19} \times \frac{1}{18}\right) =$	M1	For at least two correct triple products or fractions or decimals	ALTERNATIVE SCHEME ⁷ $C_3/^{20}C_3 + {}^6C_3/^{20}C_3 + {}^4C_3/^{20}C_3 + {}^3C_3/^{20}C_3$ M1 for at least two correct terms
		ii	$= 0.0307 + 0.0175 + 0.0035 + 0.0009$	M1	For sum of all four correct	M1 for sum of all four (all correct) either as combinations or decimals
		ii	$= 0.0526 = \frac{1}{19} \text{ (0.052632)}$	A1	CAO	A1 CAO

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$\left(= \frac{7}{228} + \frac{1}{57} + \frac{1}{285} + \frac{1}{1140} \right)$		Allow 0.053 or anything which rounds up to 0.053 with working Examiner's Comments This question was answered well (even by candidates who had struggled with earlier questions). Many answers were left as fractions (which were exact) with very few marks lost for over-specification. However many candidates squashed their work up into the first part of the answer space, not realising that there was more space on the next page. A few candidates forgot to take into consideration 'non replacement' and starting with $(7/20)^3$, gained only 1 mark, although plenty of follow through marks were available in parts (iii) and (iv).	Please check all of the answer space for this part
		iii	$P(A B) = \frac{0.0307..}{0.0526..}$	M1	For their 'A' divided by their 'B'	Allow 0.584 from $\frac{0.0307}{0.0526}$
		iii	$= 0.583 (= 0.58333)$	A1	FT their answers to (ii) provided answer < 1	$\frac{7}{12}$ Allow

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$P(B A) = 1$	B1	CAO Examiner's Comments Although over half of candidates gained at least 2 marks out of 3 here, a number of candidates did not use the straightforward $P(A B) = P(A)/P(B)$ here, and instead mistakenly calculated $P(A \cap B) = P(A) \times P(B)$, and then cancelled out a term on the top and bottom of the fraction. This illustrates the lack of deep understanding here of independence and conditionality. A disappointing number of candidates were quite happy to give an answer greater than 1 for $P(B A)$.	
		iv	$P(\text{All have orange centres})$ $= 0.0307^2 = 0.00094$ $\text{or} = \frac{49}{51984}$	M1	For their 0.0307^2	Allow 9.4×10^{-4} condone 0.0009 or 9×10^{-4}
		iv	$= (0.00094260)$	A1	FT Examiner's Comments This was well-answered although a common incorrect method was $2 \times P(A)$.	
		v	$P(\text{Has to select } > 2) = 1 - P(\text{Has to select } \leq 2)$ $= 1 - \left(\frac{14}{20} + \left(\frac{6}{20} \times \frac{14}{19} \right) \right)$ $= 1 - (0.7 + 0.221) = 1 - 0.921$	M1	For $\left(\frac{6}{20} \times \frac{14}{19} \right)$	For any of the methods below allow SC2 for $1 - 0.079 = 0.921$ or $1 - 3/38$ $= 35/38$ o.e. as final answer
		v		M1	For $1 - \text{sum of both}$	This is $1 - P(C' + CC')$
		v	$= 0.079 (=0.078947)$	A1	CAO	
		v	OR			

Question			Answer/Indicative content	Marks	Part marks and guidance	
		v	$P(\text{Has to select } > 2) =$ $P(\text{First 2 both cherry}) =$ $\left(\frac{6}{20} \times \frac{5}{19}\right)$	M2	For whole product	Without extra terms added M1 if multiplied by $k/18$ only where $0 < k < 18$ (seen as a triple product only)
		v	$= 0.079 = \frac{3}{38}$	A1	CAO	This is $P(CC)$.
		v	OR			
		v	$1 - (P(0 \text{ cherries}) + P(1$ $\text{cherry})) =$ $1 - \left(\frac{14}{20} \times \frac{13}{19} + \left(\frac{6}{20} \times \frac{14}{19}\right) + \left(\frac{14}{20} \times \frac{6}{19}\right)\right)$ $= 1 - (0.4789 + 0.2211 +$ $0.2211) = 1 - 0.9209$	M1	For any term	
		v		M1	For $1 - \text{sum of all three}$	This is $1 - P(C'C' + CC' +$ $C'C)$
		v	$= 0.079$	A1	CAO	
		v	OR			
		v	$\left(\frac{6}{20} \times \frac{5}{19} \times \frac{14}{18}\right) + \left(\frac{6}{20} \times \frac{5}{19} \times \frac{4}{18} \times \frac{14}{17}\right) +$ $\left(\frac{6}{20} \times \frac{5}{19} \times \frac{4}{18} \times \frac{3}{17} \times \frac{14}{16}\right) + \left(\frac{6}{20} \times \frac{5}{19} \times \frac{4}{18} \times \frac{3}{17} \times \frac{2}{16} \times \frac{14}{15}\right) +$ $\left(\frac{6}{20} \times \frac{5}{19} \times \frac{4}{18} \times \frac{3}{17} \times \frac{2}{16} \times \frac{1}{15} \times \frac{14}{14}\right)$	M1	For any term	This is $P(CCC' + CCCC' +$ $CCCCC' + CCCCCC' +$ $CCCCCCC')$
		v	$= \frac{7}{114} + \frac{14}{969} + \frac{7}{2584} + \frac{7}{19380} + \frac{1}{38760}$	M1	For sum of all five terms (all correct)	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		v	=0.079	A1	CAO Examiner's Comments Only approximately a quarter of candidates produced a completely correct answer, although many went through the correct answer on the way to a wrong one. Most of the correct answers used the $6/20 \times 5/19$ method, with a large minority then compromising this with another term multiplied or added.	
			Total	19		

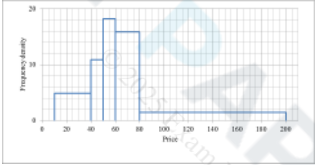
Question			Answer/Indicative content	Marks	Part marks and guidance									
7		i	<table><tr><td>0</td><td>6</td></tr><tr><td>1</td><td>5 8</td></tr><tr><td>2</td><td>1 5 8</td></tr><tr><td>3</td><td>1 1 3 5 8 9</td></tr></table> Key 1 8 represents 18 people	0	6	1	5 8	2	1 5 8	3	1 1 3 5 8 9	G1	Stem (in either order) and leaves	Do not allow leaves 21 ,25, 28 etc Ignore commas between leaves Allow stem 0, 10, 20, 30
	0	6												
	1	5 8												
2	1 5 8													
3	1 1 3 5 8 9													
	i		G1	Sorted and aligned	Allow errors in leaves if sorted and aligned. Use paper test if unsure about alignment – hold a piece of paper vertically and the columns of leaves should all be separate. Alternatively place a pencil vertically over each column. If any figures protrude then deem this as non-alignment. Highlight this error									
	i		G1	Key Examiner's Comments Most candidates scored all three marks, although some did not accurately align the leaves or did not provide a suitable key and thus scored only 2 marks. Very few candidates scored less than 2 out of 3.										
		ii	Negative	B1	Examiner's Comments This was very well-answered with only a few thinking that the skew was positive.	Allow -ve but NOT skewed to the left Do not allow 'negative correlation'								
		iii	Median = 29.5	B1	CAO									
		iii	Mean = 26.7 (26.6666) or $26\frac{2}{3}$ or $\frac{80}{3}$ or 26.6	B1	CAO	Do not allow 27 but condone 26.6 www								
		iii	Mode = 31	B1	CAO									

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	The mode is not at all useful as it is just by chance that it is 31. Mark awarded for stating not useful and -not representative of data -does not represent Central Tendency -happened by chance (or similar) -comment about not appearing significantly more (only one repetition/only twice/etc) No mark for stating it would be useful OR NOT USEFUL because of -spread/range -sample size -negatively skewed -unaffected by outliers -isn't close to mean and median	E1	Allow any reasonable comment Examiner's Comments The mean, median and mode were usually given correctly although one or two candidates lost a mark due to over-specification of the mean or rounding of the median. However the final mark for the comment was awarded to only under a quarter of candidates. Many candidates gave general descriptions of the usefulness of the mode rather than commenting on this particular case. Too many candidates stated incorrectly that the mode was useful. Those who correctly stated that it was not useful, often followed this with an incorrect reason such as being unaffected by outliers; data being negatively skewed; or not being close to the mean and/or median.	
			Total	8		

Question			Answer/Indicative content	Marks	Part marks and guidance	
8		i	Mean = $\frac{(25 \times 147) + (45 \times 109) + (55 \times 182) + (70 \times 317) + (140 \times 175)}{930}$	M1	For midpoints (at least 3 correct) (allow 25.005, 45.005 etc leading to answer 70.20)	M0A0M0A0 unless using midpoints Answer must NOT be left as improper fraction as this is an estimate
		i	$= \frac{151250}{80} =$ (£)70.19 or (£)70.2	A1	CAO (exact answer 70.19355...) Correct answers obtained from use of calculator statistical functions gain full marks Condone answer of (£)70.20	Accept correct answers for mean and sd from calculator even if eg wrong Sxx given For use of midpoints 25.5, 45.5, 55.5, 70.5, 140.5 allow SC1 for £70.69 and SC1 for 36.89
		i	$\Sigma x^2 f =$ $(25^2 \times 147) + (45^2 \times 109) + (55^2 \times 182) + (70^2 \times 317) + (140^2 \times 175)$ $= 91875 + 220725 + 550550 + 1553300 + 3430000$ $= 5846450$ $S_{xx} = 5846450 - \frac{65280^2}{930}$ $= 1264215.161 \text{ or } 5846450 - 930 \times 70.19^2$	M1	Do not FT their incorrect mean for A1	If using $(x - \bar{x})^2$ method, B2 if 36.9 or better, otherwise B0

Question			Answer/Indicative content	Marks	Part marks and guidance	
		i	$s =$ $\sqrt{\frac{1264215}{929}} = \sqrt{1360.83}$ $= 36.89$ or (£)36.9 Allow any answer between 36.87 and 36.90 without checking working	A1	(exact answer 36.88949...) Condone answer of (£)36.90 If both mean and sd overspecified, just deduct one mark Examiner's Comments This part was fairly well answered with over half of candidates gaining full credit. A few had no idea how to proceed, but most used correct midpoints, although some made slips with them or occasionally used figures such as 25.5, 45.5, etc. The standard deviation proved more difficult for a number of candidates with a variety of wrong methods seen. Very few used the statistical functions on their calculator to do this question, despite this being the recommended method. A few candidates over-specified either or both of their final answers and so lost a mark.	Allow use of 70.2 in calculation of $S_{xx} = 1263372.8$ leading to 36.87719... Condone RMSD of 36.87 (36.86985...) since n is so large

Question			Answer/Indicative content	Marks	Part marks and guidance																									
		ii	100/120 × 175 = 145.83	M1*	For 175/120 Examiner's Comments Candidates found this part rather more challenging, although almost half scored full marks. Trying to establish the proportion they were after was the biggest stumbling block. However, some were then unsure what to do with the figure of 145.83 once they had found it. Some rounded down to 145 (probably the most common mistake of those who understood what they needed to do) and others failed to finish by finding the percentage, just giving the final answer as a decimal 0.157.	Or 20/120 × 175 = 29.166 oe																								
		ii	145.83/930 = 0.1568	*M1dep		(175 – 29.166)/930																								
		ii	So 15.7%	A1		Accept 16% with working																								
		iii	<table><tr><th>Price</th><th>Frequency</th><th>Group width</th><th>Frequency density</th></tr><tr><td>10 ≤ x ≤ 40</td><td>147</td><td>30</td><td>4.90</td></tr><tr><td>40 < x ≤ 50</td><td>109</td><td>10</td><td>10.90</td></tr><tr><td>50 < x ≤ 60</td><td>182</td><td>10</td><td>18.20</td></tr><tr><td>60 < x ≤ 80</td><td>317</td><td>20</td><td>15.85</td></tr><tr><td>80 < x ≤ 200</td><td>175</td><td>120</td><td>1.46</td></tr></table>	Price	Frequency	Group width	Frequency density	10 ≤ x ≤ 40	147	30	4.90	40 < x ≤ 50	109	10	10.90	50 < x ≤ 60	182	10	18.20	60 < x ≤ 80	317	20	15.85	80 < x ≤ 200	175	120	1.46	M1	For fds - at least 3 correct Accept any suitable unit for fd such as eg freq per cm.	M1 can be also be gained from freq per 10 – 4.9, 10.9, 18.2, 15.35, 0.146 (at least 3 correct) or similar.
Price	Frequency	Group width	Frequency density																											
10 ≤ x ≤ 40	147	30	4.90																											
40 < x ≤ 50	109	10	10.90																											
50 < x ≤ 60	182	10	18.20																											
60 < x ≤ 80	317	20	15.85																											
80 < x ≤ 200	175	120	1.46																											

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii		A1	Allow 15.9 and 1.5 and condone 1.45 Examiner's Comments This part was again well answered with around 80% of candidates gaining at least 4 marks out of 5. Various errors were seen, but none very commonly. The most frequently seen were: using frequency rather than frequency density, using a non-linear scale on one of the axes (usually the horizontal axis), stopping the horizontal axis at 120, and labelling the horizontal axis 'Class width'.	If fd not explicitly given, M1 A1 can be gained from all heights correct (within $\leq$ one square) on histogram (and M1A0 if at least 3 correct)
		iii		B1	linear scales on both axes and label on both axes (Allow horizontal axis labelled x) Vertical scale starting from zero (not broken - but can get final mark for heights if broken)	Linear scale and label on vertical axis IN RELATION to first M1 mark ie fd or frequency density or if relevant $\text{freq}/10$, etc (NOT eg $\text{fd} / 10$). However allow scale given as $\text{fd} \times 10$, or similar Accept f / w or f / cw ($\text{freq} / \text{width}$ or $\text{freq} / \text{class width}$) Can also be gained from an accurate key G0 if correct label but not fd's.
		iii		B1	width of bars (within half a square) (NO GAPS ALLOWED)	Must have linear scale. Condone starting at 10 rather than 0. For inequality labels see additional notes below.
		iii	NB If not using fd's only mark available is B1 for width of bars			

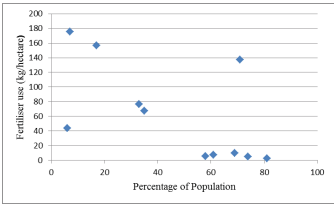
Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	Heights must be within ≤ 1 square of overlay (only for scales 2cm = 4 units (blue) or 5 units (red)) – otherwise check heights. Note that you must make sure that the overlay is aligned correctly with the vertical axis.	B1	height of bars  Examiner's Comments This part was again well answered with around 80% of candidates gaining at least 4 marks out of 5. Various errors were seen, but none very commonly. The most frequently seen were: using frequency rather than frequency density, using a non-linear scale on one of the axes (usually the horizontal axis), stopping the horizontal axis at 120, and labelling the horizontal axis 'Class width'.	Height of bars – must be linear vertical scale. FT of heights dep on at least 3 heights correct and all must agree with their fds If fds not given and 3 or 4 heights correct then max M1A0G1G1G0
		iv	Positive skewness	B1	Allow +ve Examiner's Comments Over 90% of candidates scored the one mark available here.	
		v	Area for men from 100 to 200 = $100 \times 2 = 200$	M1		Or $\frac{100}{120} \times 240$
		v	200/990 = 0.202			
		v	So 20.2%	A1		20% with working

Question			Answer/Indicative content	Marks	Part marks and guidance	
		v	Cannot be certain as both figures are estimates	E1	Independent Examiner's Comments A good number of candidates achieved full marks, and the question was answered better than question 6 part (ii) which is a similar calculation. Of those who got the calculation incorrect most started with 240/990 or 20/990, rather than 200/990. The explanation over certainty was well answered with most candidates achieving this mark, whether or not they got the first 2 marks.	Allow comments such as 'grouped data so cannot be certain' or 'Values are not exact so cannot be certain' oe or 'midpoints have been used so cannot be certain' oe
		vi	Men's running shoes have a lower average price than women's (as their mean is only £68.83 compared to £70.19). Or equivalent for women	E1	FT their mean Do NOT condone lower central tendency or lower mean	Allow 'on the whole' or similar in place of 'average'.

Question			Answer/Indicative content	Marks	Part marks and guidance	
		vi	Men's running shoes have a more variation in price than women's (as their sd is £42.93 compared to £36.89). Or equivalent for women	E1	FT their SD Examiner's Comments Although this is essentially a simple question, almost a third of candidates scored zero. Candidates struggled to provide acceptable comparisons, with many relying on terms such as "central tendency" when comparing the means, and relatively few discussing averages. A more encouraging proportion of candidates were able to provide a good interpretation for the differences in the standard deviations. Some thought that central tendency was something to do with variation. A number of candidates were unable to construct a proper, legible, grammatically correct sentence.	Allow 'more spread' or similar but not 'higher range' or 'higher variance' or 'less distributed' Condone less consistent
			Total	18		

Question			Answer/Indicative content	Marks	Part marks and guidance	
9		i	Median = 29.0	B1		Condone wrong method
		i	IQR = 31.8 – 27.8	M1	For either quartile – allow	Allow 27.75 and 31.9 leading to 4.15
		i	= 4.0	A1	alternative definitions of quartiles Examiner's Comments The vast majority of candidates found the median correctly. A small minority misread/ignored the key to the stem and leaf diagram and gave an incorrect answer of 290. However under half of candidates found the quartiles correctly, with many using 5th and 15th values, which was penalised.	Do not allow 27.7, 27.9, 31.6, 32.0
		ii	Lower limit = $27.8 - 1.5 \times 4.0 = 21.8$	M1	Method for either	For use of mean (29.44) and SD
		ii	27.75, 31.9 lead to 21.525 and 38.125	A1	FT sensible quartiles and IQR	(2.516765...) $29.44 \pm 2 \times 2.516765$ M1
			27.7, 31.6 lead to 21.85 and 37.45			Lower Limit = 24.4 A1
		ii	Upper limit = $31.8 + 1.5 \times 4.0 = 37.8$	A1	FT sensible quartiles and IQR	Upper limit = 34.5 A1 So no outliers B1

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	So there are no outliers (at either end of the distribution)	B1	Dep on at least one A1 Use of median scores 0/4 Examiner's Comments Most candidates gained full marks, often on follow through from quartiles which were slightly out. The most common error was to use the median in calculations. A few candidates started from scratch and calculated mean and standard deviation. Some managed this successfully, but others made errors in their calculations, or incorrectly used a combination of both methods such as $\text{mean} \pm 1.5 \times \text{interquartile range}$.	
			Total	7		

Question			Answer/Indicative content	Marks	Part marks and guidance	
10		i		G1	For suitably labelled axes. Condone absence of scale here.	
		i		G2,1,0	G2 for 11 points correctly plotted relative to a suitable linear scale. G1 if 9 or 10 correctly plotted. G0 if 3 or more incorrectly plotted/omitted Allow axes interchanged Examiner's Comments The scatter diagram was well drawn by many, with some choosing more manageable scales than others.	
		ii	(The points in the scatter diagram) do not appear to be roughly elliptical.	E1	For "not elliptical".	

Question			Answer/Indicative content	Marks	Part marks and guidance																																																																									
		ii	The population may not have a bivariate Normal distribution.	E1	For not underlying bivariate Normal. Do not allow “the data” in place of population/underlying. Allow “data is not from a bivariate Normal distribution”. Do not allow Normal bivariate.... Examiner's Comments The key points concerning the absence of any discernible elliptical shape and the corresponding questioning of the underlying bivariate Normal population were handled well by many. Though most candidates managed to comment on the lack of an elliptical spread of points, often with poor spelling of ‘ellipse’, there were still many candidates who struggled to differentiate between data and population.																																																																									
		iii	<table><tr><td>Percentage</td><td>33</td><td>6</td><td>58</td><td>35</td><td>81</td><td>69</td><td>61</td><td>7</td><td>74</td><td>71</td><td>17</td></tr><tr><td>Fertiliser use</td><td>76</td><td>44</td><td>6</td><td>68</td><td>3</td><td>10</td><td>7</td><td>176</td><td>5</td><td>137</td><td>157</td></tr><tr><td>Rank percentage</td><td>4</td><td>1</td><td>6</td><td>5</td><td>11</td><td>8</td><td>7</td><td>2</td><td>10</td><td>9</td><td>3</td></tr><tr><td>Rank Fertiliser</td><td>8</td><td>6</td><td>3</td><td>7</td><td>1</td><td>5</td><td>4</td><td>11</td><td>2</td><td>9</td><td>10</td></tr><tr><td>d</td><td>4</td><td>5</td><td>-3</td><td>2</td><td>-10</td><td>-3</td><td>-3</td><td>9</td><td>-8</td><td>0</td><td>7</td></tr><tr><td>d²</td><td>16</td><td>25</td><td>9</td><td>4</td><td>100</td><td>9</td><td>9</td><td>81</td><td>64</td><td>0</td><td>49</td></tr></table>	Percentage	33	6	58	35	81	69	61	7	74	71	17	Fertiliser use	76	44	6	68	3	10	7	176	5	137	157	Rank percentage	4	1	6	5	11	8	7	2	10	9	3	Rank Fertiliser	8	6	3	7	1	5	4	11	2	9	10	d	4	5	-3	2	-10	-3	-3	9	-8	0	7	d ²	16	25	9	4	100	9	9	81	64	0	49	M1	For ranking (allow ranks reversed)	
Percentage	33	6	58	35	81	69	61	7	74	71	17																																																																			
Fertiliser use	76	44	6	68	3	10	7	176	5	137	157																																																																			
Rank percentage	4	1	6	5	11	8	7	2	10	9	3																																																																			
Rank Fertiliser	8	6	3	7	1	5	4	11	2	9	10																																																																			
d	4	5	-3	2	-10	-3	-3	9	-8	0	7																																																																			
d ²	16	25	9	4	100	9	9	81	64	0	49																																																																			
		iii		M1	For d^2																																																																									
		iii	$\Sigma d^2 = 366$	A1	For Σd^2 (May be embedded in the calculation)																																																																									
		iii	$r_s = 1 - \frac{6 \sum d^2}{n(n^2 - 1)} = 1 -$ $\frac{6 \times 366}{11 \times 120} = 1 - \frac{2196}{1320}$ $= 1 - 1.6636$	M1	For method for r_s																																																																									

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$= -0.664$ (to 3 s.f.) [<i>allow</i> -0.66 to 2 s.f. or $-73/110$]	A1	FT their Σd^2 provided $-1 < r_s < 0$, and ranking used. NB No ranking scores 0/5 Examiner's Comments Many achieved full marks here. Only a few candidates reversed the ranking of one of the sets of values. Errors tended to involve mistakes in ranking, in adding the squares of the differences or in rounding the final answer. Very few candidates failed to rank the data.	
		iv	H_0 : no association between percentage of population living in rural areas and fertiliser use (in the population of countries)	B1	For null hypothesis in context NB H_0 H_1 not ρ .	
		iv	H_1 : negative association between percentage of population living in rural areas and fertiliser use (in the population of countries)	B1	For alternative hypothesis in context. Context needed in at least one of the hypotheses.	
		iv		B1	For population of countries or underlying population.	
		iv	One tail test critical value at 1% level is -0.7091	B1	For ± 0.7091 No further marks from here if incorrect.	
		iv	Since $-0.664 > -0.7091$ [or $0.664 < 0.7091$] there is ...	M1	For sensible comparison of their " -0.664 " with ± 0.7091 seen , leading to conclusion, only if $-1 < \text{their } r_s < 0$.	

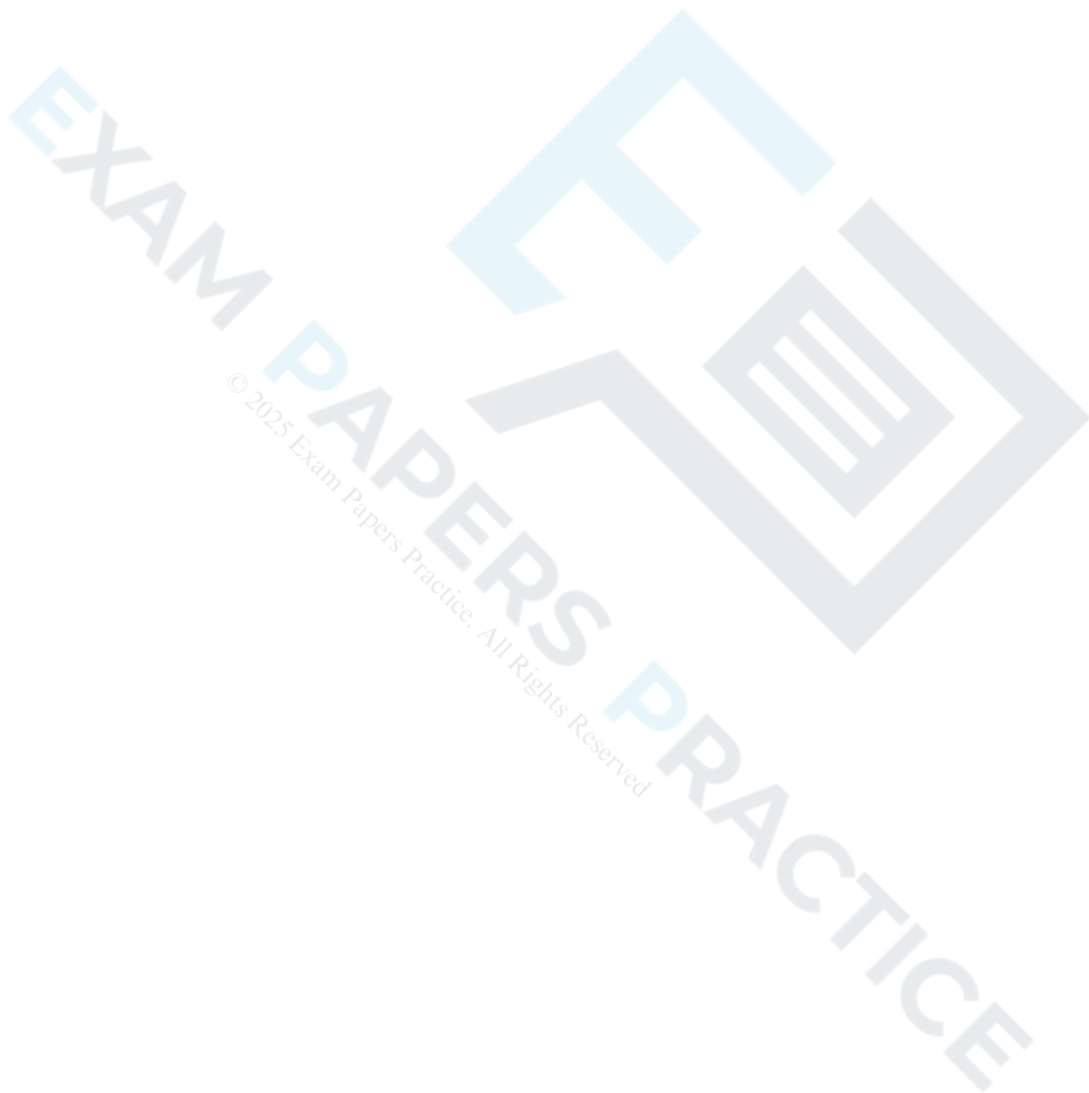
Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv	...insufficient evidence to reject H_0 . There is insufficient evidence to suggest that there is negative association between percentage of population living in rural areas and fertiliser use (in the population of countries)	A1	for not significant, oe, and correct conclusion in context. FT their r_s with correct cv. Examiner's Comments This part proved to be challenging for many candidates, especially in the identification of the underlying population of countries involved. Inappropriately worded hypotheses, using 'correlation' in place of 'association', were seen in a few cases. Many realised the one-sided nature of the test which investigated the possible negative association and went on to use the appropriate critical value to obtain a suitable conclusion in context. Few candidates expressed the conclusion in terms of the null hypothesis.	
		v	It means that the probability of rejecting H_0 given that it is correct is 1% o.e.	E1	Allow "the probability of a false positive is 1%", "the probability of a Type I Error is 1%". Do not allow "It means that the probability rejecting H_0 when it should have been accepted is 1%" Examiner's Comments There were many correct answers to this part of the question, either in terms of incorrectly rejecting H_0 or one of the equivalents. Many answers referring to 'accuracy' or 'reliability' were seen.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		vi	None	E1	Examiner's Comments	
					Few candidates appeared to understand that for this test no modelling assumptions about the underlying distribution are required.	
			Total	18		
11			Increases a value by 6 New value is closer to 62 than the old value is to 61.4 51 changes to 57 or 57 changes to 63 or 58 changes to 64	M1(AO3.1b) M1(AO2.2a) A1(AO2.2a) [3]	Implied by correct answer or pair of values differing by 6 Implied by correct answer or new value closer to 62 than old value	
			Total	3		
12		a	0.758 > 0.279 So there is sufficient evidence of correlation (in the population)	M1(AO1.1) A1(AO2.2b) [2]	Oe but not evidence of positive correlation.	
		b	E.g. diagram shows positive correlation overall, but the data consists of two distinct clusters. E.g. neither of the two clusters show evidence of correlation	B1(AO2.3) B1(AO2.2b) [2]	Accept other suitable correct comments	
			Total	4		

Question			Answer/Indicative content	Marks	Part marks and guidance		
13		a	Estimated number $= 4 + \frac{16}{3} = 9\frac{1}{3}$ $\frac{9\frac{1}{3}}{80} = 0.1166\dots$ so proportion is approximately 0.117	M1(AO3.1b) A1(AO1.1) [2]		for attempt at interpolation	
		b	E.g. Midpoints Mean Standard deviation = 3.4	M1(AO1.1) A1(AO1.1) A1(AO1.1) [3]	evidence of valid method for estimation BC Mean in the range 169-171 BC SD in the range 3-3.5		
		c	The histogram e.g. seems to have a rough bell shape e.g. is symmetrical (around the estimated mean) e.g. appears to have all data within 3 s.d. of the mean so this does support the manager's belief	B1(AO3.5a) B1(AO3.5a) [2]	for one reason for at least two reasons and 'supports belief'		
		d	i P(Lifetime > 174) for N(170, 3.4 ²) 0.1197	M1(AO3.4) A1(AO1.1)	oe BC FT their mean and standard deviation		

Question			Answer/Indicative content	Marks	Part marks and guidance	
		d	ii Answer is very similar to estimate in part (a)	B1(AO3.5a)		
				[3]		
		e	Either Test statistic $= \frac{207.3 - 210}{3.4 / \sqrt{8}} = -2.246$ Lower 5% level 1 tailed critical value of $z = -1.645$ $-2.246 < -1.645$ so significant or under $H_0, \bar{X} \sim N\left(210, \frac{3.4^2}{8}\right)$ $P(\bar{X} \leq 207.3) = 0.01235$ $0.01235 < 0.05$ so significant There is sufficient evidence to reject H_0 There is sufficient evidence to conclude that the mean lifetime is less than 210 minutes.	M1(AO3.4) A1(AO1.1) B1(AO1.1) M1(AO3.4) A1(AO1.1) B1(AO1.1) A1(AO2.2b) E1(AO2.4) [5]	Must include $\sqrt{8}$ For comparison leading to correct conclusion BC	

Question			Answer/Indicative content	Marks	Part marks and guidance
			Total	15	


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Question			Answer/Indicative content	Marks	Part marks and guidance		
14		a	Comment about shape of distribution for first graph Comment about shape of distribution for second graph	B1(AO2. 2b) B1(AO2. 2b) [2]	Comments can be combined e.g Both distributions negatively skewed gets both marks e.g. 1974 distribution has greater spread than 2014 gets both marks	If zero scored, SC1 for “The 2014 distribution is shifted to the right of the 1974 distribution” oe	
		b	i Life expectancy went down [between 1974 and 2014] in [at least] one country	E1(AO2. 2a) [1]	NOT increase in life expectancy is negative		
		b	ii The box plot is not symmetrical.	B1(AO3. 5b) [1]			
		b	iii Not appropriate with reason	E1(AO2. 4) [1]	e.g. [some] values of life expectancy are estimates The values of life expectancy are not available to this level of accuracy		
		b	iv Comment about life expectancy at birth data for countries and not individual people	B1(AO2. 4) [1]			

Question			Answer/Indicative content	Marks	Part marks and guidance	
		c	Use of $Q3 + 1.5 \times (Q3 - Q1)$ $15.873 + 1.5(8.9154) = 29.2461$ (years) The maximum value is an outlier as $30.742 > 29.2461$.	M1(AO1.2) M1(AO1.1) A1(AO1.1) [3]		
		d	<i>i</i> approx $60.8 - 37.5 = 23.3$ (years)	M1(AO3.1b)	Attempt to estimate change in life expectancy at birth soi.	
		d	<i>ii</i> Change in life expectancy for Sweden approx $81.9 - 72.5 = 9.4$ (years)	A1(AO1.1 1.1)	FT 'their 37.5 between 35 - 40'	
		d	<i>iii</i> E.g. Countries with a lower life expectancy in 1974 have greater opportunity to increase life expectancy in 2014.	A1 E1 (AO3.2a) [4]	FT 'their 72.5 between 70 - 75' OR Countries with a higher life expectancy in 1974 have less opportunity to increase life expectancy in 2014.	
		e	<i>i</i> $30.98 + 0.67 \times 37.4 = 56.0$ (years)	M1(AO3.4) A1(AO1.1) [2]		

Question			Answer/Indicative content	Marks	Part marks and guidance	
		e	ii E.g. Large amount of scatter at the lower values [and South Sudan is 37.4]. E.g. Not having the data value could indicate that there are problems in the country which could mean it does not follow the pattern for other countries	E1(AO3.5b) E1(AO3.5b) [2]	E1 Reason inferred from Fig 16.4 E1 For knowing why data may be missing	
		f	Correct method Clearly explained 6	M1(AO3.1b) E1(AO2.4) A1(AO1.1) [3]	e.g. draw “y = x” on graph e.g. The value on the vertical axis must be lower than the one on the horizontal axis FT their correct method	
			Total	20		

Question			Answer/Indicative content	Marks	Part marks and guidance		
15		a	Allocate numbers 001 to 200 to the trees Choose 10 (3 digit) random numbers	B1(AO1.2) B1(AO2.4) [2]	e.g. use calculator to get 10 different random numbers		
		b	Mean = 27.61kg SD = 4.04 kg (3sf)	B1(AO1.1) B1(AO1.1) [2]	BC BC		
		c	Upper limit = $27.61 + 2 \times 4.04 = 35.69$ So the value of 38.1 is an outlier This value should be investigated to check if it is genuine. If so, it should not be removed from the data	M1(AO1.1) A1(AO1.1) B1(AO2.2b) [3]	For mean + $2 \times \text{sd}$ OR UQ + 1.5 IQR = $28.3 + 1.5 \times 3.2 = 33.1$ OR e.g. If the value is not representative of the other 199 trees because e.g. this tree is a different type it should be ignored		
			Total	7			

Question			Answer/Indicative content	Marks	Part marks and guidance	
16		a	E.g. There is a greater spread of birth rates for countries in sub-Saharan African than for countries in the Caribbean E.g. The range for countries in Africa is greater than for countries in East and South East Asia but this could be caused by outliers as the IQRs are similar E.g. sub-Saharan Africa has a mixture of economically rich and poor countries resulting in a large IQR E.g. Countries in East and South East Asia tend to have higher life expectancy than countries in sub-Saharan Africa so their populations are older, on average, and have lower birth rates	B1(AO2. 2b) B1(AO2. 2b) B1(AO2. 2b) [3]	B1 Correct relevant comment that can be inferred from the source material B1 Distinct correct relevant comment that can be inferred from the source material B1 Third distinct relevant comment that can be inferred from the source material (this mark is only available if the candidate's comments include reference to both features of the LDS and fig 9.1)	
		b	(A) E.g. The calculation doesn't use the populations as weights E.g. Does not take the populations into account	E1(AO2. 3) [1]		
		b	(B) E.g. Lower because Australia has the highest population but the lowest birth rate oe E.g. answer given is too high as too much weight is given to Papua New Guinea	E1(AO2. 2a) [1]		

Question			Answer/Indicative content	Marks	Part marks and guidance	
		c	[weak] negative	B1(AO1.2)		
				[1]		
		d	E.g. Correlation / association does not imply causality E.g. Some countries with low birth rates have quite low physician density E.g. Some countries with low physician density have quite low birth rates E.g. Data do not show what happens after an increase in physicians Therefore it is not possible to be certain	E1(AO2.3)		
				[1]		
			Total	7		

Question			Answer/Indicative content	Marks	Part marks and guidance		
17		a	Symmetrical with one possible outlier	B1(AO1.2) [1]	or negative skew		
		b	24 th value – 8 th value 27 – 16 = 11	M1(AO1.1) A1(AO1.1) [2]			
		c	16 – 1.5 × 11 = –0.5 3 > – 0.5 so it is not an outlier.	M1(AO1.1) A1(AO2.2a) [2]	Check for outliers using their $Q_1 - 1.5 \times \text{IQR}$		
			Total	5			

Question			Answer/Indicative content	Marks	Part marks and guidance		
18		a	Consistent use of midpoints in either calculation Mean £36.25 Sample sd 8.313... £36 250 and £8313	M1(AO1.1) A1(AO1.1a) A1(AO1.1) A1(AO1.1) [4]	soi BC BC FT their calculator values		
		b	We are using grouped data not the original values	B1(AO2.4) [1]			
		c	Any valid reason which suggests that the sample is not necessarily representative	B1(AO3.2b) [1]			
		d	It would increase	B1(AO2.2a) [1]			
			Total	7			

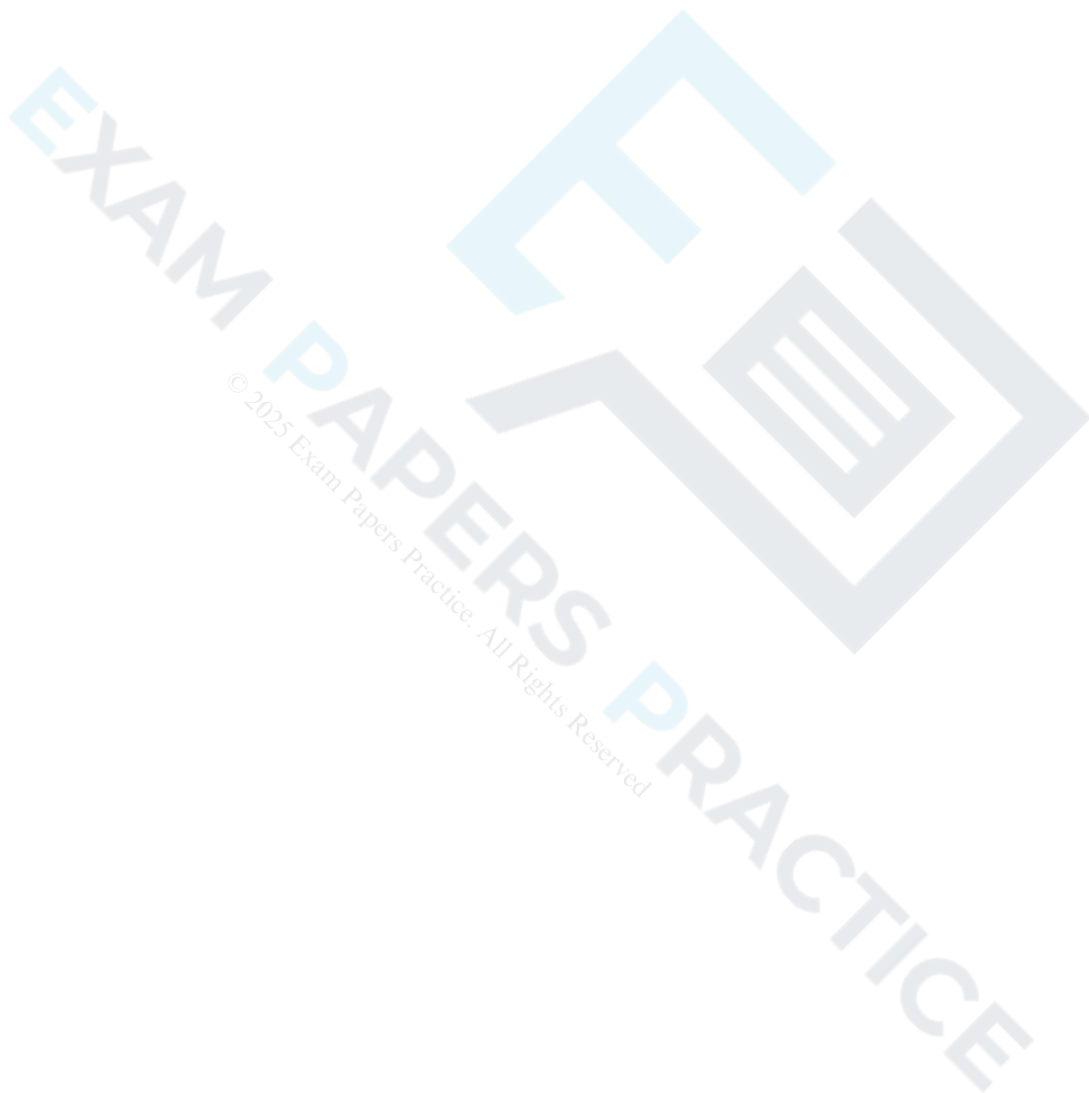
Question			Answer/Indicative content	Marks	Part marks and guidance	
19		a	Population because these are all the countries of interest.	B1(AO1.1) [1]		
		b	Eg Consistent with the correlation for all countries oe And Eg You would expect countries with higher populations to tend to have higher numbers of both mobile phone subscribers and internet users. Or Eg people who use mobile phones will be more likely to use the internet	E1(AO2.4) E1(AO2.2b) [2]		
		c	Ukraine Has high mobile phone usage with lower internet provision. Suggests people are used to and/or like technology so potential customers for the internet.	B1(AO1.1) E1(AO3.2a) [2]		

Question			Answer/Indicative content	Marks	Part marks and guidance		
		d	Number the companies from 000 to 599 or 001 to 600 Generate 3-digit random numbers (from a calculator or spreadsheet). Match number with number given to companies. [Discard 600 to 999.] Do not use any numbers twice. Stop when you have selected 20 different companies.	E1(AO1.2) E1(AO1.1) E1(AO2.4) [3]	Or from a table of random numbers. or discard 000 and 601 to 999.	OR Input list of companies into a spreadsheet NB Writing the names on paper. Putting these in a hat. Selecting 20. E1 E1E0. Not practical.	
			Total	8			
20		a	Vertical scale	B1(AO2.5) [1]			
		b	The sales are lower in the final time period (assuming a linear vertical scale) so the director is not correct.	E1(AO2.3) [1]			
			Total	2			

Question			Answer/Indicative content	Marks	Part marks and guidance		
21		a	Negative skew	B1(AO 1.2) [1]			
		b	[They are all less than 0.00005] and the table rounds values [to 4 decimal places]	E1(AO 2.4) [1]	Allow equivalent explanation in words		
		c	(18)	B1(AO 1.1) [1]			
		d	(0)	1(AO 2.2a) [1]			
			Total	4			

Question			Answer/Indicative content	Marks	Part marks and guidance		
22		a	(ai Mean for world = 7 174) $654\,290 \div 239$ = 30 million (approx.) Mean for sample is 9.36 million so much smaller	M1(AO 1.1) A1(AO 1.1) B1(AO 1.1) [3]	Any degree of accuracy		
		b	(aii There are a lot of small) countries in the world. Therefore there is no reason to suppose the sample is not random,	B1(AO 2.4) E1(AO 2.2b) [2]			
		c	(b $0.091 \times 55\,400$) =[\$] 5041.40	M1(AO 3.3) A1(AO 1.1) [2]	Condone missing units. Answer can be rounded to 5041 or 5040		
		d	(ci Identify physicians per) 1000 as highest R^2 with positive correlation and positive correlation	B1(AO 2.2b) E1(AO 2.4) [2]	NB graph C e.g. Reject D as it shows negative correlation		
		e	(cii There is hardly any) correlation.	E1(AO 3.5b) [1]			

Question			Answer/Indicative content	Marks	Part marks and guidance
			Total	10	


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Question			Answer/Indicative content	Marks	Part marks and guidance	
23		i	$\text{Mean} = \frac{17100}{50} = 342$ $S_{xx} = 6115108 - \frac{17100^2}{50} = 266908$ $s = \sqrt{\frac{266908}{49}} = \sqrt{5447.10} = 73.8 \text{ (73.8044...)}$	B1 M1 A1 [3]	Ignore units CAO For S_{xx} M1 for $6115108 - 50 \times \text{their mean}^2$ BUT NOTE M0 if their $S_{xx} < 0$ CAO ignore units M1A0 for RMSD = 73.1 (73.062...) Examiner's Comments Nearly all candidates worked out the mean correctly. Many candidates also found the standard deviation but some over-specified the answer thus losing a mark. A minority of candidates made an error in the formula for standard deviation or worked out the RMSD. It was encouraging to see most candidates recalling and using correct formulae.	

Question			Answer/Indicative content	Marks	Part marks and guidance
		ii	New mean = $(0.108 \times 342) + 7.2 = £44.14$ New sd = $0.108 \times 73.8 = £7.97$ Using RMSD gives £7.89 Using variance gives 588.29	B1 M1 A1 [3]	FT their mean Allow £44.1 or better provided answer is positive FT their sd (unless negative) for M1 and A1 NB If candidate 'starts again' only award marks for CAO Do not penalise lack of units in mean or sd Deduct at most 1 mark overall in whole question for over-specification of either mean or SD or both Examiner's Comments Most candidates used the transformation to obtain the correct mean. Many candidates also obtained the correct standard deviation, with a pleasingly small number mistakenly adding 7.2 to their final answer. Some candidates lost marks for over-specification, although there was only a penalty of 1 mark in the whole question for this error. It was encouraging to see most candidates giving answers to an appropriate degree of accuracy.
			Total	6	

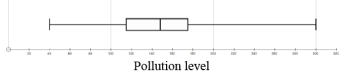
Pollutant Level x	Cumulative frequency
50	0
75	25
100	50
125	100
150	150
175	280
200	340
225	355
300	360

Question			Answer/Indicative content	Marks	Part marks and guidance	
					Examiner's Comments The majority of candidates made a good attempt at the graph. Very few failed to recognise that cumulative frequency was required, with only an occasional histogram or frequency graph seen. The values for the cumulative frequency were on the whole correctly calculated but a few tried to make them up to 365, failing to read the question correctly. The scales were usually linear but some chose difficult intervals, especially on the vertical scale, for example intervals of 24. Labelling was not as successful; missing labels or labelling the vertical scale as frequency was common. A number of candidates used mid-points rather than upper boundaries for plotting and a few used lower boundaries. Even if correct boundaries were used, the point (40,0) was often omitted with candidates either not joining their graph to the axis or joining it to (0,0). Just one third of candidates scored full marks in this question.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	Estimate from curve is 327 Proportion = $327/358 = 0.913$ or 91.3% $315/368 = 87.99\%$ $316/358 = 88.3\%$ <u>NB</u> Linear interpolation gives $284 + \frac{1}{2} \times 64 = 316$	M1 A1 [2]	Allow 315 to 330 without checking graph (unless non-linear scales in which case allow 316 by LI) Otherwise FT their graph within one square (allow a slight slip in scales – contact TL if unsure) Max M1A0 if final answer given as a fraction Examiner's Comments Candidates were fairly even spread between reading off the graph or using linear interpolation to find the cumulative frequency for $x = 200$. A sufficiently accurate value was usually obtained from the graph, but many responses stopped short of even writing it as a fraction over 358, let alone converting this value into a proportion (decimal or percentage).	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	Median = 148 Allow 145 to 152 without checking graph $Q_1 = 115$ Allow 110 to 115 without checking graph $Q_3 = 175$ Allow 175 to 180 without checking graph IQR = 60 No marks if non-linear scales If quartiles not specified give B1B0 for 'IQR is $115 < x < 175$' or similar If answer only for IQR, check if quartiles given in part (iv) or (v) – if not then check graph	B1 B1 B1 [3]	For Q_1 or Q_3 For IQR FT their cf graph for all 3 marks within one square (on both scales) (allow a slight slip in scales - contact TL if unsure) Examiner's Comments This was a generally well done. The main problems were caused by unhelpful scales chosen in part (i), which candidates then interpreted wrongly in this part. Some candidates used cumulative frequencies of 100, 200 and 300, rather than the correct values to find the median and quartiles.	
		iv	Lower limit $Q_1 - 1.5 \times \text{IQR}$ '$115 - (1.5 \times 60)$' (= 25) Upper limit $Q_3 + 1.5 \times \text{IQR}$ '$175 + (1.5 \times 60)$' (= 265) There are definitely no outliers at the lower end as the lowest data value is 40 which is below the lower limit. It is uncertain whether there	M1 M1 A1 A1 [4]	FT their quartiles provided between 40 and 300 Allow 'No values below (their) 25' for first A1 Allow 'Lower limit = (their) 25 so no outliers' You must be convinced that comments about no outliers refer to <u>lower tail only</u>. Allow <u>additional</u> comment that since some data is lost there could be one or more outliers If their lower limit > 40 then A0 Do not allow 'There <u>IS</u> at least one outlier.' oe There must be an element	

Question			Answer/Indicative content	Marks	Part marks and guidance	
			are outliers at the upper end as the highest class includes the upper limit. Use of mean = 145.08 and sd = 45.09 gives 54.9 and 235.26 for M2 So could be some outliers at lower and could be some at upper end but not sure. E1E1		of doubt. However, condone 'There is probably at least one outlier.' You must be convinced that comments about some outliers refer to upper tail only. If their upper limit <220 or >300 then A0 Examiner's Comments This was again generally well done with most candidates correctly calculating the outlier limits. Most responses correctly stated that there were no outliers at the lower end but some stated that there were definitely outliers at the upper end rather than that there may be some. A number of candidates used the median instead of the lower and upper quartiles to find the limits and others used $2 \times \text{IQR}$, rather than $1.5 \times \text{IQR}$. A very few candidates found the mean and standard deviation and then using these, found the limits correctly.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		V	 Pollution level	G1* G1*dp G1*dp [3]	FT their median and quartiles provided between 40 and 300 and $Q_1 < \text{median} < Q_3$ Can restart from graph For linear scale shown. Dep on attempt at box and whisker plot with at least a box and one whisker. Condone lack of label. For boxes (Q_1, median, Q_3) in correct positions, within half a square For whiskers at 40 and 300 within half a square Upper whisker could be partially dotted Examiner's Comments Although this part was generally answered well, a minority of candidates lost the final mark by not having the end of the whiskers plotted at 40 and/or 300, often plotting these at 29 and/or 358. Some candidates did not show a horizontal scale, making their response difficult to mark. Other candidates had trouble drawing the box and whisker diagram due the lack of a ruler.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		vi	The readings from Tower Hamlets show (stronger) positive skewness The readings from Marylebone Road show little evidence of skewness Accept 'No skewness' For 2 marks must suggest that TH has higher positive skew than MR	E1 E1 [2]	Allow '<u>slight</u> positive skewness' Do <u>not</u> FT their diagram but must have boxplot in part (v) to get second mark 'TH shows more evidence of positive skewness than MR' gets E2 Examiner's Comments Many candidates struggled to answer the question which was asked. Often zero marks were scored as the candidate wrote a short essay with no mention of skewness. Being precise and talking about both locations generally gained the marks. Some candidates still referred to it as left and right skew or mixed up positive and negative. The question did ask for a comparison, which was generally missed.	
			Total	19		

Question			Answer/Indicative content	Marks	Part marks and guidance	
25		a	$0.4a + 0.5a + 0.3a + 0.2a + 0.1a = 1$ so $a = \frac{2}{3}$	M1(AO1.1b) A1(AO1.1b) [2]		
		b	$\frac{2}{3}(0.3 + 0.2 + 0.1)$ 0.4	M1(AO1.1b) A1(AO1.1b) [2]	<i>their</i> $\frac{2}{3}$	
		c	positive skew	B1(AO1.2) [1]		
			Total	2		

Question			Answer/Indicative content	Marks	Part marks and guidance	
26		a	$= 6.8 + 1.5 \times (6.8 - 2.5) [= 13.25]$	B1(AO3.1b) [1]		
		b	Streng – large sample size th Weakn– sample is not ess random – removal of outliers may not be justifiable	B1(AO2.2a) B1(AO2.3) B1(AO2.4) [3]		
		c	eg pmcc may not be appropriate as scatter does not appear to be linear eg sample not random so calculating a correlation coefficient may not be valid eg correlation and causation are not the same thing eg the results of a hypothesis test are only suggestive, not conclusive (although highly suggestive in this case)	E1(AO2.4) E1(AO2.4) E1(AO2.4) [3]	Allow 1 mark for each distinct (sensible) reason up to a maximum of 3 	
			Total	1		

Question			Answer/Indicative content	Marks	Part marks and guidance		
27		a	eg some people have more than one phone oe	E1(AO2.4) [1]	Some of the other figures in the LDS are greater than 1		
		b	E.g. data missing E.g. removal of outlier data items	B1(AO1.1b) B1(AO1.1b) [2]			
		c	Upper quartile	B1(AO1.2) [1]			
		d	$X \sim N(996, 407^2)$, $p = 0.75$ 1271	M1(AO3.4) A1(AO1.1b) [2]	BC allow awrt 1300	if M0 allow B2 for awrt 1300 unsupported	
		e	p -value = 0.127 > 0.05 not significant Insufficient evidence to suggest that the population mean of the number of mobile phones/1000 population has increased	M1(AO1.1b) M1(AO1.1b) A1(AO2.2b) E1(AO2.2b) [4]	from screenshot comparison with 0.05 FT		

Question			Answer/Indicative content	Marks	Part marks and guidance		
		f	Sample size of 12 is small compared to the number of countries in the population so inference may be unreliable	B1(AO2.3) [1]	Any sensible comment on the premise that 12 is a relatively small sample size		
			Total	1			

Question			Answer/Indicative content	Marks	Part marks and guidance		
28		a	9×22.6 or 11×22.8 or 17×23.7 soi 203.4 , 250.8 and 402.9 isw $8 \times 8.9 + 9 \times 22.6^2$ or $10 \times 10.618 + 11 \times 22.8^2$ or $16 \times 16.9925 + 17 \times 23.7^2$ soi 4668.04 , 5824.42 and 9820.61 isw	M1(AO3.1a) A1(AO1.1b) M1(AO2.1) A1(AO1.1b) [4]	if M0 allow SC1 for use of n instead of $n - 1$ to find all three $\sum x^2$ accept answers to 4 sf or more		
		b	23.16486 to 3 sf or more $12.7(35\dots)$ to 3 sf or more	B1(AO1.1b) B1(AO1.1b) [2]	from $\frac{203.4+250.8+402.9}{37}$ from $\frac{4668.04+5824.42+9820.61-37 \times 23.1648}{36}$		
		c	Use of $N(26.1, \frac{12.735}{37})$ $26.1 \pm 1.96 \times \sqrt{\frac{12.735}{37}}$ Critical region: mean < 24.95 and mean > 27.25	M1(AO3.3) M1(AO3.1a) A1(AO3.4) [3]	FT their variance of combined sample allow use of $z = 1.645$ for M1		

Question			Answer/Indicative content	Marks	Part marks and guidance		
		d	The evidence suggests that the mean mass of adult voles has changed since the sample mean is inside the critical (rejection) region	E1(AO2.2b) [1]	or 'outside acceptance region'		
			Total	4			
29		a	Frequency density 4.6 Number of runners 21	B1(AO2.2a) B1(AO1.1) [2]			
					<u>Examiner's Comments</u> This part was usually done correctly, though some candidates omitted one or both answers. Multiples of the correct answer were sometimes seen instead of 4.6.		
		b	Vertical axis should be labelled "number of runners per minute"	E1(AO2.3) [1]	Or frequency density		
					<u>Examiner's Comments</u> The most common wrong answer was to say that the vertical axis should be labelled 'frequency'.		
			Total	3			

Question			Answer/Indicative content	Marks	Part marks and guidance	
30		a	9483	B1(AO1.1) [2]	BC <u>Examiner's Comments</u> This part was almost always done correctly.	
		b	$7 \times 10\,112 + 10259 - 10014$ soi $= 71029$ $66381 - 9204 + x = "71029"$ Emma needs to make 13852 steps	M1(AO3.1a) A1(AO1.1) M1(AO1.1) A1(AO3.2a) [4]	NB Rose's new mean is 10147 Using 8 days can gain M1 only $\frac{7 \times 10\,112 + 10259 - 10014}{7}$ $= 10147$ $= \frac{66381 - 9204 + x}{7}$ <u>Examiner's Comments</u> Most candidates found this part quite straightforward. One common error in this part was to use 10 112, the original 7-day mean for Rose, rather than work out the new 7-day mean for days 2 to 8 inclusive. Another common error was to find how many steps Emma should take on day 8 so that she has taken as many steps as Rose over the period of 8 days; this was often done by considering 8-day means.	

Question			Answer/Indicative content	Marks	Part marks and guidance		
		c	10341 + 2×948 soi = 12237 Comparison of their 13852 with their 12237 13852 is an outlier, so Emma would need to make an unusually high number of steps on day 8	M1(AO3. 1b) M1(AO1. 1) A1(AO3. 2b) [3]	Soi Conclusion; 'outlier' not essential. Dep M2 www		
					Examiner's Comments Candidates were expected to compare their answer to part (b) with the long-term mean number of steps for Emma given in the question, 10 341. In order to gain full credit candidates were expected to understand that the question was asking if the answer to part (b) was an outlier, and were expected to use the definition of outlier given in section D13 of the specification; candidates using other definitions of outlier were not given full credit. Candidates using the data in the table in the question to work out quartiles or mean and standard deviation were not given credit.		
			Total	8			

Question			Answer/Indicative content	Marks	Part marks and guidance		
31		a	Any two distinct reasons eg classes of different widths represented by bars of same width eg vertical axis should be frequency density eg final upper class boundary not given eg should have continuous horizontal scale / no gaps between bars	E1(AO2.4) E1(AO1.1) [2]	<u>Examiner's Comments</u> Many candidates gave valid answers. However, the question asked for two respects in which the presentation of the data is incorrect, and an answer like 'it should be a histogram' does not answer the question.		
		b	(i) Positive correlation (ii) 9.8395 or 9.8 or 9.84 or 9.840	B1(AO2.2b) [1] B1(AO1.1) [1]	oe <u>Examiner's Comments</u> Most candidates gave a correct answer to this part. However, stating that there is a correlation between birth and death rates was not given any credit, nor were statements like 'higher birth rates cause higher death rates'.		

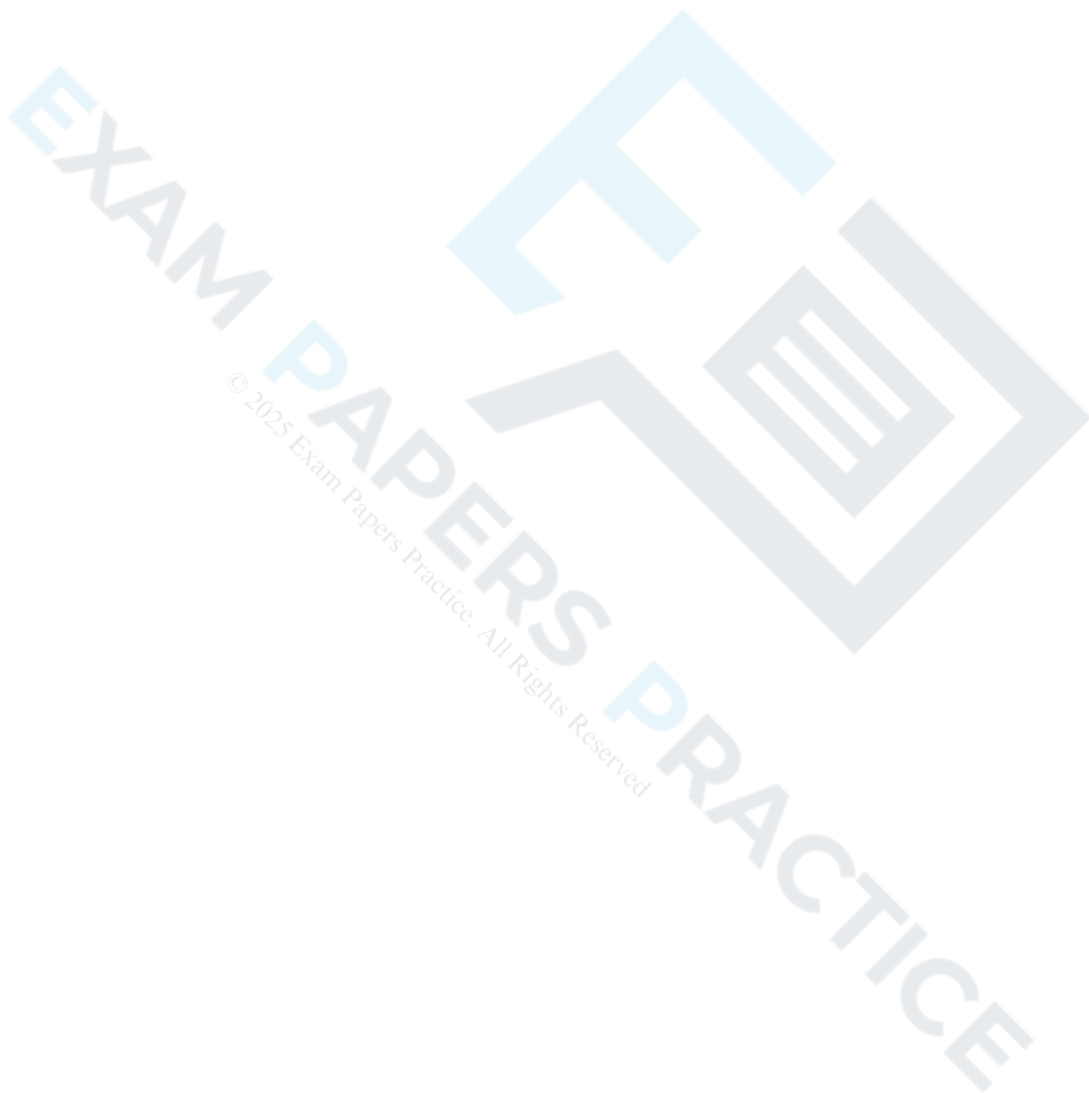
Question			Answer/Indicative content	Marks	Part marks and guidance	
			(iii extrapolation)	B1(AO3.2b) [1]	<u>Examiner's Comments</u> This question was usually answered correctly. However, a small number of candidates appeared to have read an approximate answer from the graph (usually 10) despite the clear instruction to use the equation of the line of best fit to calculate this. oe	
			(iv Birth rates and death rates in the Caribbean, may be very different from those in Africa.	E1(AO2.2a) [1]	<u>Examiner's Comments</u> Many candidates correctly answered that this would require extrapolation; many others gave an acceptable equivalent answer by pointing out that there was no data in that part of the scatter diagram so using the line of best fit would be unreliable. oe	Advantage
			(v) eg other continents to select countries from	E1(AO2.2a) [1]	<u>Examiner's Comments</u> Most candidates were sufficiently familiar with the Large Data Set to answer this part. eg a random sample would	Advantage

Question			Answer/Indicative content	Marks	Part marks and guidance		
					almost certainly not just include countries from Africa		
					<u>Examiner's Comments</u> Very few candidates answered this part correctly. Many candidates gave answers that suggested that it would be difficult to take a random sample from all the population of Africa because of communication difficulties, size of the continent, unreliability of records, etc., whilst true, do not address the context of the question. The majority of candidates did not appreciate that the sample taken was of 55 countries, and that this sample was taken from all the countries of the world, as listed in the Large Data Set.		

Question			Answer/Indicative content	Marks	Part marks and guidance	
		c	Eg Generate a random number, n, between 1 and 4 and select the nth item in the data set. Eg Select every 4th item on the list thereafter (stopping when 14 have been selected)	B1(AO1.2) B1(AO1.1) [2]	Candidates may choose other valid starting points Candidates may choose other valid intervals	Examiner's Comments There were very few completely correct answers to this part. A great many candidates described how to find a random sample by assigning random numbers to the countries. A great many others described stratified sampling. Those candidates who did describe taking, typically, every fourth country often omitted to point out the need to use a random starting point.
			Total	9		

Question			Answer/Indicative content	Marks	Part marks and guidance	
32		a	2.031578947....rounded to two or more sf isw BC	B1 (AO 1.1) [1]	NB 2.0, 2.03 or 2.032	Examiner's Comments Whilst it is acceptable to calculate the mean using a formal written method, there is an expectation in the new specification that candidates will use the statistical functions on their calculators for parts (a) and (b), hence the single mark allocation.
		b	1.076367330....rounded to two or more sf isw BC	B1 (AO 1.1) [1]	NB 1.1, 1.08 or 1.076	Examiner's Comments Candidates who did well in this question made efficient use of the appropriate calculator function. Candidates who did less well made laborious calculations and slipped up in the arithmetic.  Misconception The choice of appropriate formula (using n, $n - 1$ or other denominator corrections) is beyond the scope of this A Level Maths qualification, and the use of a single formula for all contexts is expected. The H640 OCR B (MEI) specification and formulae sheet makes clear that the divisor ($n - 1$) should be used

Question			Answer/Indicative content	Marks	Part marks and guidance
			Total	2	


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Question			Answer/Indicative content	Marks	Part marks and guidance	
33		a	negative skew	B1 (AO 1.2) [1]	<u>Examiner's Comments</u> Some confusion between positive and negative skew was seen.	
		b	(used) the mode	B1 (AO 1.1) [1]	<u>Examiner's Comments</u> A simple statement was expected, and not a mini essay. Almost every candidate gained this mark.	
		c	(used) the median	B1 (AO 1.1) [1]	<u>Examiner's Comments</u> A simple statement was expected, and not a mini essay. The majority of candidates gained the mark	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		d	$61 - 1.5 \times (88 - 61)$ $20.5 < 35$ [so 35 is not an outlier] so he does not move to set 2	M1 (AO 2.1) A1 (AO 2.2b) [2]	<i>Alternative</i> <i>y,</i> $73.61 - 2 \times 17.03$ $39.6 > 35$ [so 35 is an outlier] so he moves to set 2 allow eg only marks below 20.5 (or 39.6) would lead to a move down plus correct conclusion	Examiner's Comments Candidates who did well in this question used the lower quartile and the interquartile range to determine whether Benson's mark is an outlier. Candidates who did less well used the median in conjunction with the lower quartile.
			Total	5		

Question			Answer/Indicative content	Marks	Part marks and guidance	
34		a	(i) the cumulative frequencies have been plotted against the mid-points of the class intervals, mis-plotting [at centre of each class] reduces estimate (by 2.5) oe	B1 (AO 2.4) B1 (AO 2.4) [2]	<u>Examiner's Comments</u> Candidates who did well in this question recognised that the cumulative frequencies had been plotted at the mid-point of the intervals instead of at the upper limit.	
			(ii) grouped data has been used grouping has slightly reduced the error introduced by misplotting (because the error is less than 2.5)	B1 (AO 2.4) B1 (AO 2.4) [2]		

Question			Answer/Indicative content	Marks	Part marks and guidance	
		b	percentage unemployment is often estimated oe	E1 (AO 2.4) [1]	allow data (on percentage unemployment) is not available for all countries in Europe oe <u>Examiner's Comments</u> Candidates who did less well based comments on general geographical or economic ideas, rather than specifically related to issues related to the estimation of median values.	
		c	there are many other countries in the pre-release material; it is very unlikely that a random sample would only include European countries.	E1 (AO 2.4) [1]	<u>Examiner's Comments</u> Candidates who did well on part (b) and part (c) were familiar with the pre-release material made appropriate comments.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		d	negative correlation / association (may be embedded) comparison of p -value with 0.05 or 0.01 or other appropriate significance level and supporting comment	B1 (AO 2.2b) B1 (AO 2.2b) [2]	if B0B0 allow SC2 for eg comment on no significant association justified by comparison of p -value with appropriate significance level (eg 0.025)	

Examiner's Comments

Candidates who did well commented on the nature of the correlation. They compared the p -value with a significance level and then made an appropriate deduction. Candidates who did less well compared the correlation coefficient with the p -value or did not comment on the association at all.

Question			Answer/Indicative content	Marks	Part marks and guidance	
		e	(even though this is interpolation), the scatter / weak correlation / presence of an outlier would suggest that the use of a line of best fit is inappropriate	E1 (AO 2.2b) [2]	allow explanation based on the value for Kosovo being an outlier or on it lying in the (large) gap in the scatter <u>Examiner's Comments</u> Candidates who did well commented on the nature of the scatter or the position of 35.3 relative to the given values to justify their comment.	
			Total	9		
35		a	Positive skew	B1 (AO1.2) [1]		
		b	Layout correct scale on axis and range of boxplot correct their IQR and median shown IQR is 19 to 40 and median is 28	M1 (AO1.1a) A1 (AO1.1) M1 (AO1.1) A1 (AO1.1) [4]	allow 20.5 to 41 but not 19 to 41 or 20.5 to 40	
			Total	5		

Question			Answer/Indicative content	Marks	Part marks and guidance	
36		a	the pattern of data points suggest a straight line or the value of r indicates that the fit is close or	E1 (AO3.3) E1 (AO2.4) [2]		
		b	70	B1 (AO3.4) [1]	from $0.89 \times 83 - 3.76 (= 70.11)$	
		c	$(10 + 3.76) \div 0.89$ 13	M1 (AO3.4) A1 (AO1.1) [2]	(= 13.21 ...)	
		d	The approximation for Tina's mark is obtained by interpolation, whereas the approximation for Dave's mark is from extrapolation	E1 (AO3.5b) [1]	allow eg should use the equation for x on y to estimate Dave's mark	
			Total	6		
37		a	$=B2 \times C2$	E1 (AO2.4) [3]	must have " $=$ "	
		b	A All populations are in the upper quartile B All total GDPs are in the upper quartile Both statements are consistent with the data.	E1 (AO2.4) E1 (AO2.4) B1 (AO2.2a) [3]		
		c	No because some countries do not take part in the Olympics OR having more Olympic athletes in a country may encourage others	E1 (AO2.2a) [1]		
			Total	5		

Question			Answer/Indicative content	Marks	Part marks and guidance		
38		a	0.6	B1 (AO1.1) [1]	BC		
		b	each egg has the same (constant) probability of being cracked	E1 (AO2.4) [1]	OR the probability of any particular egg being cracked is independent of any other egg being cracked or		
		c	$np = \text{their } 0.6$ $p = 0.1$	M1 (AO3.1b) A1 (AO1.1) [2]			
		d	159.4, 106.3, 29.5, 4.4, 0.4, 0.0, 0.0	M1 (AO3.4) M1 (AO1.1) A1 (AO1.1) [3]	Use of B(6, 0.1) so Multiplication by 300 All correct	If unsupported, allow B3 for all frequencies correct, B2 if one error, B1 if two errors	
		e	The theoretical frequencies are close to the observed values, so Rose's model is a good fit.	B1 (AO3.2b) [1]			

Question			Answer/Indicative content	Marks	Part marks and guidance		
		f	$H_0: p = 0.1$ $H_0: p \neq 0.1$ p is the probability that an egg selected at random is cracked use of $B(24, 0.1)$ $P(X \leq 5) = 0.9723 \dots BC$ $1 - 0.9723 > 0.025$ not significant oe There is no evidence at the 5% level to suggest that the probability that an egg selected at random is cracked is not 0.1	B1 (AO1.1) B1 (AO2.5) M1 (AO3.3) A1 (AO1.1) M1 (AO3.4) A1 (AO1.1) E1 (E12.2b) [7]	Both hypotheses Comparison of their $1 - p(x \leq 5)$ with 0.025		
			Total	15			

Question			Answer/Indicative content	Marks	Part marks and guidance	
39		a	Opportunity sampling	B1(AO 1.2) [1]		
		b	Each sample of this size does not have an equal probability of being selected since not all the customers will come at 11.00 am on a Monday	E1(AO 2.4) [1]	allow eg Some of Qasim's customers will be at work and never be able to visit café at 11am on Monday	
		c	Positive skew	B1(AO 1.2) [1]		
		d	Median is 10 th value = 30 43 – 22 = 21	B1(AO 1.1) M1(AO 1.1) A1(AO 1.1) [3]	For their 15 th value – their 5 th value	NB other conventions for finding the quartiles are acceptable as long as the method is clear.
		e	Any two reasonable statements eg Medians similar, so age of typical customer found to be the same IQR much smaller in larger sample, suggesting less variability range is larger, (but these values are both outliers and could be atypical)	B1(AO 2.4) B1(AO 2.4) [2]		
			Total	8		

Question		Answer/Indicative content	Marks	Part marks and guidance		
40		$\bar{X} = 130$ seen $H_0 : \mu = 142$ $H_1 : \mu < 142$ μ is the population mean length of time of meetings use $N\left(142, \frac{17.4^2}{9}\right)$ of $P(\bar{X} < 130) = 0.01927...$ $0.01927... < 0.05$ So [reject H_0], there is evidence at the 5% level to suggest that the mean length of meeting time has decreased.	B1 (AO 1.1) B1(AO 1.1) B1(AO 2.5) M1(AO 3.3) A1(AO 1.1) M1(AO 3.4) A1(AO 2.2b) [1]	BC soi BC FT comparison of their probability		
		Total	7			

Question			Answer/Indicative content	Marks	Part marks and guidance	
41		a	If a particular data value is not available, the code #N/A has been included in the Large Data Set – this is to prevent some software reading a blank as a zero.	E1 (AO 2.4) [1]		
		b	The populations of African countries vary considerably. The “mean of means” does not take weighting into account.	E1 (AO 2.4) [1]		
		c	The heights of the bars are proportional to the frequencies in the frequency diagram, the areas of the bars are proportional to frequency in a histogram (or the heights of the bars are proportional to frequency density)	E1 (AO 2.4) [1]		
		d	Frequency densities of 1.3, 2.4, 3.8 and 0.733 Horizontal and vertical scales correct and correctly labelled Correct diagram with no gaps between bars	B1 (AO 1.1) B1 (AO 1.1) B1 (AO 1.1) [3]		
		e	eg there may some association between the variables, but it is not clear what sort, so using a value of one variable to predict a value of the other variable is unlikely to be reliable	B1 (AO 2.4) B1 (AO 2.2b) [2]	or eg in the subsection of scatter where median life expectancy is 17, there appears to be no correlation	
			Total	8		

Question			Answer/Indicative content	Marks	Part marks and guidance	
42	a		2.8 to 2.81 BC	B1 (AO1.1) [1]	NB 2.8063 583815029 .. Examiner's Comments Candidates who did well in this question used their calculator efficiently in this part – there was no need to show any working. Those that did attempt this part long hand often made arithmetic errors.	
	b		eg the data is already grouped oe eg we do not have the original raw data oe eg we are using the mid-point of the intervals eg we are assuming the data are uniformly distributed across each interval oe	B1 (AO2.4) [1]	Examiner's Comments In this part only about half the candidates were able to explain clearly why their answer to the part above is an estimate, with vague or partially correct explanations seen.	
	c		eg using upper class limit in each case gives mean is 3.4...so it is possible that mean is more than 3	B1 (AO2.3) [1]	or eg need 520 or more and using the upper limits gives 590 simply stating eg the mean could be 3.4 is insufficient Examiner's Comments Candidates that did well in this part recognised that the request to “determine” necessitated providing some supporting calculations.  AFL Questions involving the interpretation of a	

Question			Answer/Indicative content	Marks	Part marks and guidance
					mathematical model should be answered with reference to the context of the question and not simply using generic textbook descriptions. Short calculations that support an argument should be encouraged, rather than mini-essays. Exemplar 1 If all times were at the top of the class: $\frac{\sum x}{n} = \frac{590}{173} = 3.41 > 3 \therefore \text{is possible}$ Exemplar 1 shows a clear justification for full credit in this part. Exemplar 2 The estimated mean is 2.81 so it is possible that the actual mean hours spent answering emails is over 3 hours if, in each group of jobs, most of the values are above the midpoints of each group used to estimate the mean. Exemplar 2 shows insufficient justification for full credit in this part. The candidate has shown a general understanding of the disadvantages of working with grouped data but has not applied this knowledge explicitly in the context of the data given in this question.
			Total	3	

Question			Answer/Indicative content	Marks	Part marks and guidance	
43	a		2.5	B1 (AO1.1) [1]	<u>Examiner's Comments</u> Candidates who did well in this question were familiar with the terminology and gave concise and correct answers. Candidates who did less well calculated the median instead of the mid-range in part (a).	
	b		positive skew cao	B1 (AO1.1) [1]	<u>Examiner's Comments</u> The majority of candidates correctly identified that the distribution had a positive skew.	
	c		1	B1 (AO1.1) [1]	<u>Examiner's Comments</u> The majority of candidates correctly stated the mode value.	
			Total	3		

Question		Answer/Indicative content	Marks	Part marks and guidance	
44	a	the data was not available for all countries oe	B1 (AO2.4) [1]	<u>Examiner's Comments</u> Candidates who did well in this question commented that the data may not have been available in the other three countries. Candidates who did not do well were over-imaginative in their responses, making comments on the countries being outliers or at war, but not linking this explicitly with the result that the data was not available.	
	b	use of $Q_1 - 1.5 \times (Q_3 - Q_1)$ and $Q_3 + 1.5 \times (Q_3 - Q_1)$ seen for either set $4.135 < 6.28$ and $15.775 > 14.46$ $0.38 < 3.58$ and $18.86 > 14.89$	M1 (AO3.1b) A1 (AO1.1) A1 (AO1.1) [3]	if A0A0 allow SC1 for 4.135, 15.775, 0.38 and 18.86 all seen	<u>Examiner's Comments</u> Candidates who did well used the criteria for outliers based on the interquartile range and compared their results with the given smallest and largest values. Those who did less well made a slip with the arithmetic or did not make any specific comparison. Candidates who did not do well tried to work with the mean and standard deviation or were unable to use the criteria based on interquartile range, median $\pm 1.5 \times \text{IQR}$ being a common error.

Question			Answer/Indicative content	Marks	Part marks and guidance		
	c		22 954 isw	B1 (AO3.1b) [1]	allow 22 955, 22 950 or 23 000	NB $6411776 \times \frac{3.58}{1000}$	
					<u>Examiner's Comments</u> Candidates who did well were familiar with the LDS and understood that crude death rate is quoted per thousand. They gave their answer as a suitable whole number. Candidates who did not do well assumed the crude death rate was per 100 people or per day.		
	d		there are almost certainly more "old" people in the population of	B1 (AO2.4) [1]	<u>Examiner's Comments</u> Candidates who did well commented on the higher proportion of older people in Germany. Candidates who did badly commented how people might die a crude death, that Germany has a larger population or guessed at, for example, higher crime rates.		
	e		in African countries there is a negative association / relationship between (or negative correlation between the ranks of) median age and crude death rate, but in Europe there seems to be a positive association / relationship between (or positive correlation between the ranks of) median age and crude death rate	B1 (AO2.4) B1 (AO2.4) [2]	do not allow "negative correlation" and / or "positive correlation"	comment comparing and contrasting type of relationship in both continents for B1, and one comment comparing and contrasting	

Question			Answer/Indicative content	Marks	Part marks and guidance	
			the “association” / “relationship between” or “correlation between the ranks of” median age and crude death rate (appears to be) stronger in Europe		allow B1 both relationship s are weak oe strength of relationship in both continents for B1 allow equivalent explanation s in words eg as median age increases crude death rates decrease in Africa and similar for Europe <u>Examiner's Comments</u> Candidates who did well in this question understood the difference between association and correlation and commented accordingly. They commented on the nature of the association and the relative strength of the association for each continent. Candidates who did not do well were in the majority. They commented on correlation and/or the strength of the linear relationship. Some candidates did not really address the question at all, instead writing commentaries on the different social and economic conditions in the two continents.	
			Total	8		

Question		Answer/Indicative content	Marks	Part marks and guidance		
45	a	Correct structure for two boxplots seen Range correct and clear on one diagram IQR correct and clear on one diagram Median correct and clear on one diagram Both boxplots completely correct with all details shown as above	B1 (AO1.1a) B1 (AO1.1) B1 (AO1.1) B1 (AO1.1) B1 (AO1.1)	F: 82.1 to 86.7 M: 77.6 to 83.3 F: 83.45 to 84.9 M 79 to 81.15 F: 84 M: 80.25 F/M label or with key	Through template circles	
			[5]	<u>Examiner's Comments</u> Almost all candidates were able to draw two boxplots. However, only a minority were completely correct, with many errors in plotting values.		

Question		Answer/Indicative content	Marks	Part marks and guidance	
	b	80.25 < 84 so typical male has lower life expectancy than typical female 2.15 > 1.45 OR 5.7 > 4.6 so life expectancy of males is more variable	E1 (AO2.2a) E1 (AO2.2a) [2]	OR 80.2844 < 84.2313 so the average life expectancy is lower for males OR 1.4294 > 1.1563 so life expectancy of males is more variable Or median (or mean) life expectancy for males is less than for females Or life expectancy for males is more variable than for females + ref to range, IQR or sd	Examiner's Comments Candidates were expected to make comments in context and provide support from the information given in the question for their comment. While many did this well, quite a lot of candidates either only quoted values from the information given in the question or only made an unsupported comment – like females live longer than males. A good answer is shown in Exemplar 2. Exemplar 2 On average women have a higher life expectancy as the median for women is 84 > 80.25 for men. Women have a more consistent set of results for life expectancy as their interquartile range of 1.45 < 2.15 for men, therefore there is less of a range of variation in life expectancy.

Question			Answer/Indicative content	Marks	Part marks and guidance		
	c		eg The LDS deals with averages, not individuals, so it does not mean that everyone in London has a longer life expectancy than everyone in Lancashire eg There is no info available for Lancashire	E1 (AO2.4) E1 (AO2.4) [2]	Any two valid different comments <u>Examiner's Comments</u> Many candidates appeared to have little knowledge of the large data set, judging by common answers like 'there is no data for Barnet alone in the large data set'. The most common correct answer was that there was no information available in the large data set for Lancashire alone, although some candidates did not appear to know that Lancashire is not a London borough. A common incorrect answer was that the summary statistics were for the range 2012-2014, and so would not apply to 2013.		

Question			Answer/Indicative content	Marks	Part marks and guidance	
	d		Either M: $80.2844 \pm 2 \times 1.4294$ ($= 83.1432$) or F: $84.2313 \pm 2 \times 1.1563 = 86.5439$ $83.1(432) < 83.3$ and $86.5(439) < 86.7$	M1 (AO1.1) A1 (AO2.4) [2]	Correct use of criterion Both correct comparisons <u>Examiner's Comments</u> There were many good answers to this question, with some candidates explaining carefully that there could be more than two outliers, but there were certainly at least two. Some candidates did not use the definition for outliers based on the mean as standard deviation, which the question required; others did not know the definition involved 2 standard deviations. A small number of candidates found the correct boundaries for outliers, but then failed to compare the boundaries correctly with both maximum values.	83.1(432) or 81.9(187) or 86.5(439) or 77.4(256)
	e		(linear) correlation (strong) positive	M1 (AO2.2b) A1 (AO1.1) [2]	<u>Examiner's Comments</u> Most candidates provided the expected correct answer of positive correlation.	
			Total	13		

Question			Answer/Indicative content	Marks	Part marks and guidance		
46	a		Diagram A is incorrect the cumulative frequencies have been plotted at the left hand end of each class interval, not at the right hand end	B1 (AO2.2a) B1 (AO2.3) [2]	Accept diagram B is correct Oe, eg Used lower bound not upper bound		
					<u>Examiner's Comments</u> The majority of candidates wrongly answered that Diagram B is incorrect, giving a variety of false reasoning. Follow through was allowed for part (b) so these candidates could gain credit for a reading from the wrong diagram; however, many candidates did not read very accurately, and some only got as far as identifying the 49th value.		
	b		35.7 to 35.9 inclusive	B1 (AO1.1) [1]	From diagram, but ignore any calculation; allow even if wrong diagram given in (a).	If they say Diagram B is incorrect oe in (a), allow sc B1 for 34.7 to 34.9 inclusive	
			Total	3			

Question			Answer/Indicative content	Marks	Part marks and guidance	
47	a		Quota sampling	B1 (AO1.2) [1]	Examiner's Comments Very few knew that the question was describing Quota sampling, though many realised it could result in bias through not being a random sample. A good many candidates knew that the answer to part (c) is 'Positive skew', though a considerable number of them described the diagram as having a peak at 1 and then tailing off. Although most candidates could find the mean in part (d), wrong answers, like 16.67, were common and few could find the standard deviation. Only a minority of candidates answered part (e) correctly. Candidates did better on part (f), partly because follow through from their answers to part (d) was allowed, although some did not put their comments in context.	
	b		non-random	B1 (AO2.4) [1]	Or equivalent	Accept 'the procedure is biased', but not 'those chosen might be biased' or 'tedious'. Ignore extra parts of explanation.
	c		Positive skew	B1 (AO1.2) [1]		

Question			Answer/Indicative content	Marks	Part marks and guidance		
	d		mean is 1.67 sd is 1.31	B1 (AO1.1) B1 (AO1.1) [2]	CAO CAO Do not allow 1.30 or 1.3		
	e		20	B1 (AO1.1) [1]	NB $44 \times \frac{100}{220}$		
	f		(means very similar so) average number of portions of fruit consumed per child is similar to average number of portions of fruits consumed per adult (sd for children is lower so) less variability in number of portions of fruit consumed per day by children	B1 (AO2.2a) B1 (AO2.5) [2]	Condone mean for children is (slightly) less than for adults BUT must be in context of portions of fruit	ft from (d); answers must be consistent with their answers to (d) and must both be in context	
			Total	8			

Question		Answer/Indicative content	Marks	Part marks and guidance	
48	a	Mass = 23.56×1.816^2 = 77.7 [kg]	M1 (AO1.1) A1 (AO1.1) [2]	Allow M1 for cm used here CAO	Examiner's Comments Most candidates were able to find the mass correctly, though some did not realise the height is squared in the formula, and others used the height in cm rather than m. The correlation in part (b) was generally described correctly as 'positive' or by a description such as 'as age increases so does BMI'. Part (c) was usually done correctly, but few gave the answer that (d) would require extrapolation, which is not sensible. Some candidates did correctly explain that the data only went up to an age of 45, and so the line of best fit could not be used at 60, while others gave an incomplete argument, that the lack of data at (or about) age 60 meant the line of best fit should not be used. For part (e), a good number of candidates correctly pointed out that the sample data included no females, and the range of ages used was restricted. There were no completely correct answers to part (f); some candidates gained a mark for saying that every nth value should be taken (where n was usually 16), but none explained the need for a random starting point.

Question			Answer/Indicative content	Marks	Part marks and guidance		
	b		Positive [correlation]	B1 (AO1.1) [1]	Ignore, eg, weak or strong	Accept 'as age increases so does BMI' oe	
	c		$28 \leq \text{BMI} < 30$	B1 (AO3.4) [1]			
	d		extrapolation	B1 (AO3.5b) [1]	Or equivalent		
	e		eg females in population wider age range in population scatter diagram for whole population shows very weak negative correlation	B1 (AO2.2b) B1 (AO2.2b) [2]	any two distinct comments		
	f		Eg Generate a random number, n , between 1 and (eg 20) inclusive and select the n th item in the data set OR select every m th (eg 16th value). For valid solution.	M1 (AO1.2) A1 (AO1.1) [2]	Random no between 1 and 9 and every 17 th item would also work.	Random no between 1 and 31, and then every 15 th item also works, etc.	
			Total	9			

Question			Answer/Indicative content	Marks	Part marks and guidance	
49	a		$23 \leq m \leq 29$	B1 (AO1.1) [1]		
	b		no, $p\%$ spent less than an hour revising maths; or no 90% spent less than m minutes revising maths	B1 (AO2.4) [1]	$75 \leq p \leq 85$ $75 \leq m \leq 100$	Allow eg x out of 200 is not 90% or 0.9 oe where $150 \leq x \leq 170$
					Examiner's Comments Candidates who did well in Question 4 were able to provide a realistic estimate of the median in part (a) and went on to make a correct statement in part (b) which was supported by a correct argument. Candidates who did less well provided insufficient detail to support their (usually correct) statement in part (b), or mistakenly thought that the median time was 60 minutes.	
			Total	2		

Question			Answer/Indicative content	Marks	Part marks and guidance	
50	a		Quota sampling	B1 (AO1.2) [1]		
	b		9	B1 (AO1.1) [1]	from 5×1.8	
	c		Systematic: select every 24th number on the list start randomly between $n = 1$ and $n \geq 184$ and stop when 200 have been selected (if $n > 184$, must cycle through list) Simple random sampling: assign each item in the list a unique number (eg from 1 to 4960) generate random numbers until a sample of 200 has been selected so i	M1 (AO2.4) A1 (AO1.1) E1 (AO2.4) E1 (AO1.1) [4]	<i>alternatively</i> select every 25th number on the list start randomly between $n = 1$ and $n \geq 25$, and cycle through the list again, stopping when 200 have been selected <i>alternatively</i> allow any process where each member of the population has an equal chance of being selected allow any	<i>alternatively</i> select every 24.8th value on list, rounding as appropriate start randomly with any value on list. Cycle through the list repeatedly until 200 items have been selected

Question			Answer/Indicative content	Marks	Part marks and guidance		
					process where each possible sample has an equal chance of being selected		
	d		as the size of the sample increases, the shape of the distribution appears more and more "Normal" oe	B1 (AO2.4) [1]	must refer to shape and closer to Normal shape for larger sample		
	e		use of $N(60.0515, 6.5783^2)$ to find $P(X > 65)$ awrt 0.23 $4960 \times \text{their } 0.226$ 1121 or 1120 or 1119	M1 (AO3.3) A1 (AO3.4) M1 (AO3.1b) A1 (AO3.5a) [4]	condone use of 6.5717^2 or parameters rounded to 3 sf	M0 if continuity correction used or eg $P(X > 64)$ found	
	f		eg there may be seasonal fluctuations such as teachers retiring in August	B1 (AO3.5b) [1]	allow any sensible reason in context	do not allow eg mean and sd may be different	
			Total	12			

Question			Answer/Indicative content	Marks	Part marks and guidance	
51	a		house prices are generally higher in London boroughs (than elsewhere in the country), so Dr Procter's suggestion is probably wrong	B1 (AO2.2a) [1]		
	b		214 505 219 402	B1 (AO3.4) B1 (AO1.1) [2]		
	c		$P = 28\,500Y - 57\,184\,000$ (where Y is the calendar year) or $P = 28\,500y + 215\,000$ (where y is the number of years after 2014)	B1 (AO3.3) B1 (AO1.1) [2]	gradient intercept	allow both marks for correct equation in any form isw allow eg $y = 28\,500x - 57\,184\,000$
	d		2016 272 000 2017 300 500	B1(AO3.4) B1(AO1.1) [2]		FT <i>their</i> straight line model provided this gives values > 250 000
	e		Dr Procter's model is a (very) poor fit Prof Jackson's is a good fit, or works well for 2017, but not 2016	B1 (AO2.2a) B1 (AO2.2a) [2]	dependent on correct values in (b) FT comment for <i>their</i> values > 250 000	this mark is dependent on having calculated values in part (d)
	f		neither – extrapolation oe	B1 (AO3.5b) [1]		
			Total	10		

Question		Answer/Indicative content	Marks	Part marks and guidance		
52	a	$2p + q + 0.2 + 0.3 = 1$ soi oe $2 \times p \times q = 0.06$ soi eliminate p or q with a correct substitution from one of <i>their</i> equations $q^2 - 0.5q + 0.06 = 0$ or $2p^2 - 0.5p + 0.03 = 0$ oe $q = 0.2$ or 0.3 and $p = 0.15$ or 0.1 $(q < 2p \text{ so}) q = 0.2$ and $p = 0.15$	B1 (AO2.1) M1 (AO3.1a) M1 (AO1.1) A1 (AO1.1) A1 (AO1.1) A1 (AO3.2a) [6]	allow M1 if 2 omitted eg $2 \times \frac{0.03}{q} + q = 0.5$ or $2p + \frac{0.03}{p} = 0.5$ may be implied by eg $q = 0.2$ or 0.3 and $2p = 0.3$ or 0.2	NB if 2 omitted, A0 for $2p^2 - 0.5p + 0.6 = 0$ or $2q^2 + q + 0.24 = 0$ which have no real root	
	b	$10 \times q \times (1 - q)^9$ soi 0.27 or 0.268 or awrt 0.2684 isw	M1 (AO1.1) A1 (AO1.1) [2]	FT <i>their</i> q where $0 < q < 1$		

Question		Answer/Indicative content	Marks	Part marks and guidance		
	c	$H_0: p = 0.2$ $H_1: p > 0.2$ p is the probability that the spinner shows a 1 (on any given spin) oe use of $X \sim B(100, 0.2)$ where x is the number of 1s obtained in 100 spins to obtain $P(X \geq k)$ or $P(X \leq k)$ $P(X \leq 27) = \text{awrt } 0.97$ or $P(X \geq 28) = \text{awrt } 0.034$ $0.034 < 0.05$ or $0.97 > 0.95$ significant or reject H_0 or accept H_1; may be embedded in conclusion in context there is sufficient evidence to suggest (at 5% level) that the probability of a score of 1 is greater than 0.2	B1 (AO1.1) B1 (AO2.5) M1 (AO3.3) A1 (AO1.1) M1 (AO3.4) A1 (AO1.1) A1 (AO2.2b) [7]	both hypotheses ; allow equivalent in words or eg $P(1) = 0.2$ $k = 27, 28$ or 29 or critical region is $X \geq 28$ or 28 is in critical region must have the correct probability or correct critical region for the last two A marks	allow any parameter as long as clearly defined as probability M0 for $P(X = k)$ NB $P(X = 28) = 0.014...$ $P(X = 27) = 0.020168...$ FT <i>their</i> probability, dependent on award of first M1 do not allow eg conclude / prove / indicate or other assertive statement instead of suggest; A0 if answer spoiled	
		Total	15			

Question			Answer/Indicative content	Marks	Part marks and guidance		
53	a		Lee is wrong because he should make the comparison of 0.033 with 0.05 he should make the comparison of 0.37154 with 0	B1 (AO2.2a) B1 (AO2.2b) [2]	allow he should have compared r with the critical value	if B0B0 SC1 for Lee has confused r with p or for 0.37154 suggests positive correlation	
	b		$465467 + 2 \times 204356$ awrt 874180 (or 867940 from use of 210236) from scatter diagram the outliers are approximately 920 000, 1 200 000	M1 (AO2.1) A1 (AO2.2b) [2]	condone use of 201236 instead of 204356; ignore work relating to lower tail numerical values must be mentioned	or $521000 + 1.5 \times (521000 - 342500)$ or 788750 in which case accept two or three outliers identified extra one is approximately 800 000	
	c		the pmcc would (probably) be closer to 0 because the scatter is less well modelled by a straight line the p-value would increase because a value which is closer to 0 is more likely assuming there is no correlation	B1 (AO2.2b) B1 (AO2.2b) [2]	if B0B0 allow SC1 for r closer to 0 and p-value larger		
	d		the student's suggestion is reasonable, since there are other regions defined in the LDS	B1 (AO2.2b) [1]			
			Total	7			